

SingleRAN

Base Station Cabinets and Subracks (Including the BBU Subrack) Configuration Feature Parameter Description

Issue

Draft A

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 01 (2025-03-10).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added the configurations of the BTS5900R cabinet and subrack. For details, see: • 3.1.1.1 Mapping Between Base Station Models and Cabinet Types • 3.1.1.2 Cabinet Configuration Principles • 3.1.3.1 Subrack Types • 3.1.3.2 Subrack Configuration Principles • 3.1.4 BBU Board Types • 3.1.6.6 BTS5900R Base Station • 3.5.1.2 Using MML Commands • 3.5.4.5.1 Resetting BBU Boards	None	5900 series base stations (macro base stations)
Optimized the configuration for the number of ports on the CPRI panel. For details, see 3.5.4.3 Modifying BBU Board Configuration .	Added the CASCADEPORT.PORTU SAGE (LTE eNodeB, 5G gNodeB) parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Deleted the UTRPb4 board.	None	3900 series base stations (macro base stations)

Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 Introduction

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements of the operating environment that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR. This feature works the same way with each of these RATs.

3 Base Station Cabinets and Subracks

This document describes which cabinets and subracks are required for different base station models and how to configure them. This document also describes how to configure and maintain boards in a baseband unit (BBU).

3.1 Principles

Huawei provides 3900 and 5900 series base stations to suit different scenarios, enabling quick and cost-effective network deployment. Main base station hardware includes cabinets, BBUs, and RF modules. For details about Huawei base stations, see *Base Station Hardware* in *3900 & 5900 Series Base Station Product Documentation*.

3.1.1 Introduction to 5900 Series Base Stations and Cabinets

3.1.1.1 Mapping Between Base Station Models and Cabinet Types

5900 series base stations have the following models:

BTS5900, BTS5900L, BTS5900A, BTS5929R, DBS5900, DBS5900A, and DBS5900E

5900 series base stations support the following cabinet types:

BTS5900 (Ver.A), BTS5900L (Ver.A), BTS3900 (Ver.E_A~D), BTS3900L (Ver.E_B~D), APM30H (Ver.E), APM5930, APM5900H, APM5950H, APM5950H-L, APM5900H-L, TMC11H (Ver.E), RFC, RFC5906 series cabinet, IBBS200D/T, IBBS300D/T, BBC5200D/T, BBC5300D/T, BBC5200D-L, BBC5600D, BBC5600D-V, BBC5600A, BBC5600A-V, ILC29 (Ver.E), IMB05, IFS5906 series rack, OPM200, OBU5900A, OBU5900C, VIRTUAL, OBU5901, BTS5900R

For details on the mapping between base station models and cabinet types, see the following tables.

 NOTE

- **VIRTUAL** indicates a virtual cabinet. It can be an object in physical entity modeling, for example, a non-Huawei legacy cabinet or a cabinet type not included in existing cabinet types. It can also be an object when there is no physical entity, for example, a base station installed in an open rack or independent module such as an OPM50M that has no installed cabinets.
- The hyphen (-) indicates N/A.

Table 3-1 Mapping between BTS5900 base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
BTS3900 cabinet	BTS3900 (Ver.E_A~D)	Indoor Macro Middle Cabinet, Ver.D	BTS3900
BTS5900 cabinet	BTS5900 (Ver.A)	BTS5900, Ver.A	BTS5900
Virtual cabinet	-	-	VIRTUAL

Table 3-2 Mapping between BTS5900L base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
BTS3900L cabinet	BTS3900L (Ver.E_B~D)	Indoor Macro Large Cabinet, Ver.D	BTS3900L
BTS5900L cabinet	BTS5900L (Ver.A)	BTS5900L, Ver.A	BTS5900L
Virtual cabinet	-	-	VIRTUAL

Table 3-3 Mapping between BTS5900A base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
APM30 series cabinet	APM30H (Ver.E)	APM30H, Ver.E	APM30
TMC series cabinet	TMC11H (Ver.E)	TMC11H, Ver.E	TMC

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
RFC5906 series cabinet	RFC5906 (Ver.A)	RFC5906, Ver.A	RFC5906
Battery cabinet	IBBS200D (Ver.C)/ IBBS200T (Ver.C)	IBBS200D/IBBS200T, 303V.C	BBC
	IBBS200D (Ver.D)/ IBBS200T (Ver.D)	IBBS200D/IBBS200T, Ver.D	
	IBBS200D (Ver.E)/ IBBS200T (Ver.E)	IBBS200D/IBBS200T, Ver.E	
	IBBS300D (Ver.A)/ IBBS300T (Ver.A)	IBBS300D/IBBS300T	
	BBC5200D (Ver.A)/ BBC5200T (Ver.A)	BBC5200D/BBC5200T, Ver.A	
	BBC5200D-L	BBC5200D-L, Ver.A	
	BBC5300D (Ver.A)/ BBC5300T (Ver.A)	BBC5300D/BBC5300T, Ver.A	
	BBC5600D (Ver.A)	BBC5600D, Ver.A	
	BBC5600D-V (Ver.A)	BBC5600D-V, Ver.A	
	BBC5600A (Ver.A)	BBC5600A, Ver.A	
	BBC5600A-V (Ver.A)	BBC5600A-V, Ver.A	
RFC cabinet	RFC (Ver.C)	RFC, 303V.C	RFC
	RFC (Ver.D)/RFC (Ver.D_A)/RFC (Ver.D_B)	RFC, Ver.D	
	RFC (Ver.E)/RFC (Ver.E_A)/RFC (Ver.E_B)	RFC, Ver.E	

Table 3-4 Mapping between BTS5929R base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
BTS5929R cabinet	-	-	BTS5900R

Table 3-5 Mapping between DBS5900 base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
Virtual cabinet, including ILC29 (Ver.E), virtual cabinet (Ver.D), and others	-	-	VIRTUAL
IMB05	-	-	BTS3900D
IBC10(DC)	IBC10(DC) Ver.B	IBC10 DC, Ver.B	IBC10
APM30 series cabinet	APM30H (Ver.E)	APM30H, Ver.E	APM30
APM5930 cabinet	APM5930(AC) (Ver.A)	APM5930(AC), Ver.A	APM5930
	APM5930(DC) (Ver.A)	APM5930(DC), Ver.A	
APM5950 cabinet	APM5950H	APM5950H, Ver.A	APM5950
	APM5950H-L	APM5950H-L, Ver.A	
APM5900 cabinet	APM5900H	APM5900H, Ver.A	APM5900
APM5900 cabinet	APM5900H-L	APM5900H-L, Ver.A	APM5900
TMC series cabinet	TMC11H (Ver.E)	TMC11H, Ver.E	TMC
RFC5906 series cabinet	RFC5906 (Ver.A)	RFC5906, Ver.A	RFC5906
IFS5906 series rack	-	IFS5906, Ver.A	IFS5906
Outdoor blade power module OPM200	-	-	OPM
Battery cabinet	IBBS200D (Ver.C)/ IBBS200T (Ver.C)	IBBS200D/IBBS200T, 303V.C	BBC
	IBBS200D (Ver.D)/ IBBS200T (Ver.D)	IBBS200D/IBBS200T, Ver.D	
	IBBS200D (Ver.E)/ IBBS200T (Ver.E)	IBBS200D/IBBS200T, Ver.E	

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
	IBBS300D (Ver.A)/ IBBS300T (Ver.A)	IBBS300D/IBBS300T	
	BBC5200D (Ver.A)/ BBC5200T (Ver.A)	BBC5200D/BBC5200T, Ver.A	
	BBC5300D (Ver.A)/ BBC5300T (Ver.A)	BBC5300D/BBC5300T, Ver.A	
	BBC5200D-L (Ver.A)	BBC5200D-L, Ver.A	
	BBC5600D (Ver.A)	BBC5600D, Ver.A	
	BBC5600D-V (Ver.A)	BBC5600D-V, Ver.A	
	BBC5600A (Ver.A)	BBC5600A, Ver.A	
	BBC5600A-V (Ver.A)	BBC5600A-V, Ver.A	

Table 3-6 Mapping between DBS5900A base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
OBU5900A cabinet	-	-	OBU5900 A
Outdoor blade power module OPM200	-	-	OPM

Table 3-7 Mapping between DBS5900E base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
OBU5901 cabinet	-	-	OBU5901

Table 3-8 Mapping between DBS5900E base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
OBU5900C cabinet	-	-	OBU5900C

The USU3910 (data exchange unit) can be used for interconnecting BBUs.

Table 3-9 Mapping between USU3910 base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
IBC10(DC)	IBC10(DC) Ver.B	IBC10 DC, Ver.B	IBC10

3.1.1.2 Cabinet Configuration Principles

Cabinet numbers range from 0 to 62, and cabinets where BBUs reside must be numbered 0 to 7. Otherwise, software cannot work properly. The following tables describe the configuration principles.

Table 3-10 Cabinet configuration principles for the BTS5900 and BTS5900L

Logical Name of the Cabinet	Cabinet Number	Remarks
BTS3900/BTS3900L/ BTS5900/BTS5900L/ VIRTUAL	0 to 7	Numbered according to the sequence of deploying base stations. If a site has only one BBU, the BBU is configured in cabinet 0; if a site has two BBUs, it is recommended that the primary BBU be configured in cabinet 0.
Reserved	8 or 9	Reserved
APM30/TMC/ VIRTUAL/BBC	10 to 19	Remote cabinets used for monitoring UMTS and LTE RRUs, such as power supply cabinets
APM30/TMC/ VIRTUAL/BBC	20 to 29	Remote cabinets used for monitoring GSM RRUs, such as power supply cabinets

Logical Name of the Cabinet	Cabinet Number	Remarks
Reserved	30 to 62	Others

Table 3-11 Cabinet (including outdoor APM30) configuration principles for the BTS5900A and DBS5900

Logical Name of the Cabinet	Cabinet Number	Remarks
APM30/TMC	0 to 7	APM30, TMC, or virtual cabinet where a BBU resides, numbered according to the deployment sequence. If a site has only one BBU, the BBU is configured in cabinet 0; if a site has two BBUs, it is recommended that the primary BBU be configured in cabinet 0.
RFC/RFC5906	1 to 4	RFCs where RFUs reside, or RFC5906 cabinets where RRUs reside
APM30/TMC/VIRTUAL	5 to 7	APM30, TMC, or virtual cabinets
OPM	5 or 6	Outdoor blade power module OPM200
TMC	8	Transmission cabinets (with no BBUs configured)
BBC	9 to 14	APM30-supported battery cabinets
APM30/TMC/VIRTUAL/BBC/OMB	10 to 19	Remote cabinets used for monitoring UMTS/LTE RRUs, such as power supply cabinets (In new deployment scenarios, these cabinet numbers are preferentially used for BBCs.)
APM30/TMC/VIRTUAL/BBC/OMB	20 to 29	Remote cabinets used for monitoring GSM RRUs, such as power supply cabinets
Reserved	30 to 62	Others

Table 3-12 Cabinet (including outdoor APM5930) configuration principles for the DBS5900

Logical Name of the Cabinet	Cabinet Number	Remarks
APM5930	0 to 7	Cabinets where BBUs reside, numbered according to the deployment sequence. If a site has only one BBU, the BBU is configured in cabinet 0; if a site has two BBUs, it is recommended that the primary BBU be configured in cabinet 0.
RFC/RFC5906	1 to 4	RFCs where RFUs reside, or RFC5906 cabinets where RRUs reside
APM5930/VIRTUAL	5 to 7	APM5930 or virtual cabinets
APM5930	8	APM5930(DC) transmission cabinets (with no BBUs installed)
OPM	5 or 6	Outdoor blade power module OPM200
BBC	9 to 14	APM5930-supported battery cabinets
APM5930/ VIRTUAL/BBC/OMB	10 to 19	Remote cabinets used for monitoring RRUs, such as power supply cabinets (In new deployment scenarios, these cabinet numbers are preferentially used for BBCs.)
Reserved	30 to 62	Others

Table 3-13 Cabinet (including outdoor APM5950 or APM5900) configuration principles for the DBS5900

Logical Name of the Cabinet	Cabinet Number	Remarks
APM5950	0 to 7	Cabinets where BBUs reside, numbered according to the deployment sequence. If a site has only one BBU, the BBU is configured in cabinet 0; if a site has two BBUs, it is recommended that the primary BBU be configured in cabinet 0.

Logical Name of the Cabinet	Cabinet Number	Remarks
RFC/RFC5906	1 to 4	RFCs where RFUs reside, or RFC5906 cabinets where RRUs reside
APM5900	8	APM5900 transmission cabinets (with no BBUs installed)
OPM	5 or 6	Outdoor blade power module OPM200
BBC	9 to 14	APM5950-supported battery cabinets
Reserved	30 to 62	Others

Table 3-14 Cabinet (including outdoor OBU5900A) configuration principles for the DBS5900A

Logical Name of the Cabinet	Cabinet Number	Remarks
OBU5900A	0 to 7	Cabinets where the BBU5900A resides, numbered according to the deployment sequence. If a site has only one BBU, the BBU is configured in cabinet 0; if a site has two BBUs, it is recommended that the primary BBU be configured in cabinet 0.
OPM	5 or 6	Outdoor blade power module OPM200

Table 3-15 Cabinet configuration principles when indoor USU3910 modules are used

Logical Name of the Cabinet	Cabinet Number	Remarks
IBC10	0 to 7	Cabinets where the USU3910 resides, numbered according to the deployment sequence

Table 3-16 Cabinet (including outdoor OBU5901) configuration principles for the DBS5900E

Logical Name of the Cabinet	Cabinet Number	Remarks
OBU5901	0 to 7	Cabinets where BookBBU5901 resides, numbered according to the deployment sequence. It is recommended that the BBU at the site be configured in cabinet 0.

Table 3-17 Cabinet (including outdoor OBU5901) configuration principles for the DBS5900E

Logical Name of the Cabinet	Cabinet Number	Remarks
OBU5901	0 to 7	Cabinets where the BookBBU5901/ BookBBU5901a resides, numbered according to the deployment sequence. It is recommended that the BBU at the site be configured in cabinet 0.

Table 3-18 Cabinet (including outdoor OBU5900C) configuration principles for the DBS5900E

Logical Name of the Cabinet	Cabinet Number	Remarks
OBU5900C	0 to 7	Cabinets where the BBU5900C resides, numbered according to the deployment sequence. If a site has only one BBU, the BBU is configured in cabinet 0; if a site has two BBUs, it is recommended that the primary BBU be configured in cabinet 0.

Table 3-19 Cabinet (including outdoor BTS5900R) configuration principles for the BTS5929R

Logical Name of the Cabinet	Cabinet Number	Description
BTS5900R	0 to 7	Integrated cabinets, numbered according to the deployment sequence. It is recommended that the BBU at the site be configured in cabinet 0.

3.1.2 Introduction to 3900 Series Base Stations and Cabinets

3.1.2.1 Mapping Between Base Station Models and Cabinet Types

3900 series base stations have the following models:

BTS3900, BTS3900L, BTS3900A, BTS3900AL, DBS3900, and BTS3900C

3900 series base stations support the following cabinet types:

BTS3900, BTS3900L, APM series cabinet, RFC, RFC5906 series cabinet, TMC series cabinet, OMB series cabinet, battery cabinet, BTS3900AL, TP48600A-H17B1, BTS3012 series cabinet, IFS5906 series rack, outdoor blade power module OPM200, VIRTUAL, and others

For details on the mapping between base station models and cabinet types, see the following tables.

 NOTE

VIRTUAL indicates a virtual cabinet. It can be an object in physical entity modeling, for example, a non-Huawei legacy cabinet or a cabinet type not included in existing cabinet types. It can also be an object when there is no physical entity, for example, a base station installed in an open rack or independent module such as an OPM50M, BBU3910A, or BBU3910C that has no installed cabinets.

The hyphen (-) indicates N/A.

Table 3-20 Mapping between BTS3900 base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
BTS3900 cabinet	BTS3900 (Ver.C)	Indoor Macro Middle Cabinet, 103V.C	BTS3900
	BTS3900 (Ver.D)/BTS3900 (Ver.D_A)/BTS3900 (Ver.D_B)	Indoor Macro Middle Cabinet, Ver.D	BTS3900
BTS3012 cabinet	BTS3012 (Ver.D_Z)	BTS3012	BTS3012_S-RAN
BTS3012II/ BTS3012III cabinet	BTS3012 (Ver.D_Z)	BTS3012	BTS3012II_S-RAN
Virtual cabinet	-	-	VIRTUAL

Table 3-21 Mapping between BTS3900L base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
BTS3900L cabinet	BTS3900L (Ver.C)	Indoor Macro Large Cabinet, 202V.C	BTS3900L
	BTS3900L (Ver.D)/ BTS3900L (Ver.D_B)	Indoor Macro Large Cabinet, Ver.D	
Virtual cabinet	-	-	VIRTUAL

Table 3-22 Mapping between BTS3900A base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
APM series cabinet	APM30H (Ver.C)	APM30H, 303V.C	APM30
	APM30H (Ver.D)/APM30H (Ver.D_A1)/APM30H (Ver.D_A2)/APM30H (Ver.D_B)	APM30H, Ver.D	
	APM30H (Ver.E)/APM30H (Ver.E_A1)/APM30H (Ver.E_A2)/APM30H (Ver.E_B)	APM30H, Ver.E	
TMC series cabinet	TMC11H (Ver.C)	TMC11H, 303V.C	TMC
	TMC11H (Ver.D)/TMC11H (Ver.D_A1)/TMC11H (Ver.D_A2)/TMC11H (Ver.D_B)	TMC11H, Ver.D	
	TMC11H (Ver.E)/TMC11H (Ver.E_A1)/TMC11H (Ver.E_A2)/TMC11H (Ver.E_B)	TMC11H, Ver.E	
Battery cabinet	BBC	-	BBC
	IBBS200D (Ver.C)/IBBS200T (Ver.C)	IBBS200D/IBBS200T, 303V.C	
	IBBS200D (Ver.D)/IBBS200T (Ver.D) IBBS200D (Ver.D_A)/IBBS200T (Ver.D_A) IBBS200D (Ver.D_B)/IBBS200T (Ver.D_B)	IBBS200D/IBBS200T, Ver.D	
	IBBS200D (Ver.E)/IBBS200T (Ver.E) IBBS200D (Ver.E_A)/IBBS200T (Ver.E_A) IBBS200D (Ver.E_B)/IBBS200T (Ver.E_B)	IBBS200D/IBBS200T, Ver.E	
	IBBS300D (Ver.A)/IBBS300T (Ver.A)	IBBS300D/IBBS300T	
	IBBS20D	IBBS20D	
	BBC5200D-L	BBC5200D-L, Ver.A	

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
	IBBS200D-L (Ver.F_D~E)/ IBBS200T-L (Ver.F_D~E)	IBBS200D-L/IBBS200T-L, Ver.F	
	IBBS700D/IBBS700T	IBBS700D/IBBS700T	
RFC5906 series cabinet	RFC5906 (Ver.A)	RFC5906, Ver.A	RFC5906
RFC cabinet	RFC (Ver.C)	RFC, 303V.C	RFC
	RFC (Ver.D)/RFC (Ver.D_A)/RFC (Ver.D_B)	RFC, Ver.D	
	RFC (Ver.E)/RFC (Ver.E_A)/RFC (Ver.E_B)	RFC, Ver.E	
BTS3012AE cabinet	BTS3012AE (Ver.D_Z)	BTS3012AE	BTS3012A E_SRAN
Virtual cabinet	-	-	VIRTUAL

Table 3-23 Mapping between BTS3900AL base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
BTS3900AL cabinet	BTS3900AL (Ver.A)	BTS3900AL BTS3900AL	BTS3900A L
TMC series cabinet	TMC11H (Ver.E)	TMC11H, Ver.E	TMC
Battery cabinet	IBBS300D (Ver.A)/ IBBS300T (Ver.A)	IBBS300D/IBBS300T BBC	BBC
	IBBS700D/IBBS700T	IBBS700D/IBBS700T BBC	BBC
Virtual cabinet	-	- VIRTUAL	VIRTUAL

Table 3-24 Mapping between DBS3900 base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
APM series cabinet	APM30H (Ver.C)	APM30H, 303V.C	APM30
	APM30H (Ver.D)/APM30H (Ver.D_A1)/APM30H (Ver.D_A2)/APM30H (Ver.D_B)	APM30H, Ver.D	
	APM30H (Ver.E)/APM30H (Ver.E_A1)/APM30H (Ver.E_A2)/APM30H (Ver.E_B)	APM30H, Ver.E	
APM5930 cabinet	APM5930(AC) (Ver.A)	APM5930(AC), Ver.A	APM5930
	APM5930(DC) (Ver.A)	APM5930(DC), Ver.A	
APM5950 cabinet	APM5950H	APM5950H, Ver.A	APM5950
	APM5950H-L	APM5950H-L, Ver.A	
APM5900 cabinet	APM5900H	APM5900H, Ver.A	APM5900
APM5900 cabinet	APM5900H-L	APM5900H-L, Ver.A	APM5900
TMC series cabinet	TMC11H (Ver.C)	TMC11H, 303V.C	TMC
	TMC11H (Ver.D)/TMC11H (Ver.D_A1)/TMC11H (Ver.D_A2)/TMC11H (Ver.D_B)	TMC11H, Ver.D	
	TMC11H (Ver.E)/TMC11H (Ver.E_A1)/TMC11H (Ver.E_A2)/TMC11H (Ver.E_B)	TMC11H, Ver.E	
Battery cabinet	BBC	-	BBC
	IBBS200D (Ver.C)/IBBS200T (Ver.C)	IBBS200D/IBBS200T, 303V.C	
	IBBS200D (Ver.D)/IBBS200T (Ver.D) IBBS200D (Ver.D_A)/IBBS200T (Ver.D_A) IBBS200D (Ver.D_B)/IBBS200T (Ver.D_B)	IBBS200D/IBBS200T, Ver.D	

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
	IBBS200D (Ver.E)/ IBBS200T (Ver.E) IBBS200D (Ver.E_A)/ IBBS200T (Ver.E_A) IBBS200D (Ver.E_B)/ IBBS200T (Ver.E_B)	IBBS200D/IBBS200T, Ver.E	
	IBBS200D-L (Ver.F_D~E)/ IBBS200T-L (Ver.F_D~E)	IBBS200D-L/IBBS200T-L, Ver.F	
	IBBS300D (Ver.A)/ IBBS300T (Ver.A)	IBBS300D/IBBS300T	
	IBBS20D	IBBS20D	
	IBBS700D/IBBS700T	IBBS700D/IBBS700T	
RFC5906 series cabinet	RFC5906 (Ver.A)	RFC5906, Ver.A	RFC5906
IFS5906 series rack	-	IFS5906, Ver.A	IFS5906
Outdoor blade power module OPM200	-	OPM200, Ver.A/Ver.B	OPM
TP48600A-H17B1 cabinet	TP48600A-H17B1	TP48600A	TP48600 A
Virtual cabinet, including the indoor centralized rack (ICR), 19-inch rack, OPM50M, BBU3910A, BBU3910C, and others	-	-	VIRTUAL
IMB03	-	-	BTS3900 D
IBC10(DC)	IBC10	IBC10 DC, Ver.A	IBC10
	IBC10(DC) Ver.B	IBC10 DC, Ver.B	IBC10
OMB cabinet	OMB (Ver.C)	OMB	OMB

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
MRE1000	MRE1000	MRE1000	MRE1000

Table 3-25 Mapping between BTS3900C base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
OMB cabinet	BTS3900C/BTS3900C (Ver.C)	OMB	OMB

 NOTE

The BTS3900C is a type of distributed base station. The logical names of other cabinets used by the BTS3900C are configured to the same as those of the DBS3900.

The USU3910 (data exchange unit) can be used for interconnecting BBUs.

Table 3-26 Mapping between USU3910 base station and cabinet types

Physical Cabinet	Cabinet Nameplate	Key Words on Electronic Labels of Cabinets	Logical Name
IBC10(DC)	IBC10(DC) Ver.B	IBC10 DC, Ver.B	IBC10
IBC10(DC)	IBC10(DC) Ver.B	IBC10 DC, Ver.B, LG U +	IBC10

3.1.2.2 Cabinet Configuration Principles

Cabinet numbers range from 0 to 62, and cabinets where BBUs reside must be numbered 0 to 7. Otherwise, software cannot work properly. The following tables describe the configuration principles.

Table 3-27 Cabinet configuration principles for the BTS3900 and BTS3900L

Logical Name of the Cabinet	Cabinet Number	Remarks
BTS3900/ BTS3900L/ VIRTUAL	0 to 7	Cabinets where BBUs reside, numbered according to the deployment sequence. If a site has only one BBU, the BBU is configured in cabinet 0; if a site has two BBUs, it is recommended that the primary BBU be configured in cabinet 0.
Reserved	8 or 9	Reserved
APM30/TMC/ VIRTUAL/BBC/OM B	10 to 19	Remote cabinets used for monitoring UMTS and LTE RRUs, such as power supply cabinets
APM30/TMC/ VIRTUAL/BBC/OM B	20 to 29	Remote cabinets used for monitoring GSM RRUs, such as power supply cabinets
Reserved	30 to 62	Others

Table 3-28 Cabinet configuration principles for the BTS3900AL

Logical Name of the Cabinet	Cabinet Number	Remarks
BTS3900AL/ VIRTUAL	0 to 7	Cabinets where BBUs reside, numbered according to the deployment sequence. If a site has only one BBU, the BBU is configured in cabinet 0; if a site has two BBUs, it is recommended that the primary BBU be configured in cabinet 0.
TMC	8	Transmission cabinet
BBC	9 or 10	BTS3900AL-supported battery cabinets
APM30/TMC/ VIRTUAL/BBC/OM B	10 to 19	Remote cabinets used for monitoring UMTS and LTE RRUs, such as power supply cabinets (In new deployment scenarios, these cabinet numbers are preferentially used for BBCs.)
APM30/TMC/ VIRTUAL/BBC/OM B	20 to 29	Remote cabinets used for monitoring GSM RRUs, such as power supply cabinets
Reserved	30 to 62	Others

Table 3-29 Cabinet (including outdoor APM30) configuration principles for the BTS3900A and DBS3900

Logical Name of the Cabinet	Cabinet Number	Remarks
APM30/TMC	0	APM30, TMC, or virtual cabinet where a BBU resides, numbered according to the deployment sequence. If a site has only one BBU, the BBU is configured in cabinet 0; if a site has two BBUs, it is recommended that the primary BBU be configured in cabinet 0.
RFC/RFC5906	1 to 4	RFCs where RFUs reside, or RFC5906 cabinets where RRUs reside
APM30/TMC/VIRTUAL	5 to 7	APM30, TMC, or virtual cabinets
TMC	8	Transmission cabinets (with no BBUs configured)
BBC	9 to 14	APM30-supported battery cabinets
APM30/TMC/VIRTUAL/BBC/OMB	10 to 19	Remote cabinets used for monitoring UMTS/LTE RRUs, such as power supply cabinets (In new deployment scenarios, these cabinet numbers are preferentially used for BBCs.)
APM30/TMC/VIRTUAL/BBC/OMB	20 to 29	Remote cabinets used for monitoring GSM RRUs, such as power supply cabinets
Reserved	30 to 62	Others

Table 3-30 Cabinet (including outdoor APM5930) configuration principles for the DBS5900

Logical Name of the Cabinet	Cabinet Number	Remarks
APM5930	0 to 7	Cabinets where BBUs reside, numbered according to the deployment sequence. If a site has only one BBU, the BBU is configured in cabinet 0; if a site has two BBUs, it is recommended that the primary BBU be configured in cabinet 0.
RFC/RFC5906	1 to 4	RFCs where RFUs reside, or RFC5906 cabinets where RRUs reside

Logical Name of the Cabinet	Cabinet Number	Remarks
APM5930/VIRTUAL	5 to 7	APM5930 or virtual cabinets
APM5930	8	APM5930(DC) transmission cabinets (with no BBUs installed)
OPM	5 or 6	Outdoor blade power module OPM200
BBC	9 to 14	APM5930-supported battery cabinets
APM5930/ VIRTUAL/BBC/OMB	10 to 19	Remote cabinets used for monitoring RRUs, such as power supply cabinets (In new deployment scenarios, these cabinet numbers are preferentially used for BBCs.)
Reserved	30 to 62	Others

Table 3-31 Cabinet (including outdoor APM5950 or APM5900) configuration principles for the DBS5900

Logical Name of the Cabinet	Cabinet Number	Remarks
APM5950	0 to 7	Cabinets where BBUs reside, numbered according to the deployment sequence. If a site has only one BBU, the BBU is configured in cabinet 0; if a site has two BBUs, it is recommended that the primary BBU be configured in cabinet 0.
RFC/RFC5906	1 to 4	RFCs where RFUs reside, or RFC5906 cabinets where RRUs reside
APM5900	8	APM5900 transmission cabinets (with no BBUs installed)
OPM	5 or 6	Outdoor blade power module OPM200
BBC	9 to 14	APM5950-supported battery cabinets
Reserved	30 to 62	Others

Table 3-32 Indoor cabinet, OMB cabinet, or MRE1000 cabinet configuration principles for the DBS3900

Logical Name of the Cabinet	Cabinet Number	Remarks
OMB/VIRTUAL/ MRE1000	0 to 7	Virtual cabinets, OMB cabinets, or MRE1000 cabinets where BBUs reside, numbered according to the sequence of deploying base stations
IFS5906	1 to 4	IFS5906 racks where RRUs reside
OPM	5 or 6	Outdoor blade power module OPM200
Reserved	8 or 9	Reserved
APM30/TMC/ VIRTUAL/BBC/OMB	10 to 19	Remote cabinets used for monitoring UMTS and LTE RRUs, such as power supply cabinets
APM30/TMC/ VIRTUAL/BBC/OMB	20 to 29	Remote cabinets used for monitoring GSM RRUs, such as power supply cabinets
Reserved	30 to 62	Others

Table 3-33 Cabinet configuration principles for the DBS3900 in outdoor scenarios where BBU3910A modules are used

Logical Name of the Cabinet	Cabinet Number	Remarks
VIRTUAL	0	Virtual cabinet where a BBU resides and virtual cabinet where the first OPM50M connecting to the BBU resides
VIRTUAL	5 to 7	Virtual cabinets where other OPM50M modules connecting to the BBU reside
OPM	5 or 6	Outdoor blade power module OPM200 or virtual cabinets
BBC	9 to 14	Auxiliary battery cabinets
VIRTUAL	Others	OPM50M modules that monitor RRUs
BBC	Others	IBBS20D cabinets that monitor RRUs

Table 3-34 Cabinet configuration principles for the DBS3900 in outdoor scenarios where BBU3910C modules are used

Logical Name of the Cabinet	Cabinet Number	Remarks
VIRTUAL	0	Virtual cabinet where a BBU resides

Table 3-35 Cabinet configuration principles for upgraded BTS3012 series base stations

Logical Name of the Cabinet	Cabinet Number	Remarks
BTS3012_SRAN/ BTS3012AE_SRAN	0 to 7	Numbered according to the sequence of deploying base stations. Cabinets where BBUs reside are numbered from 0 to 7. If a site has only one BBU, the BBU is configured in cabinet 0; if a site has two BBUs, it is recommended that the primary BBU be configured in cabinet 0.
Reserved	8	Reserved
BBC	9 to 12	Auxiliary battery cabinets
APM30/TMC/ VIRTUAL/BBC/OM B	10 to 19	Remote cabinets used for monitoring UMTS/LTE RRUs, such as power supply cabinets (In new deployment scenarios, these cabinet numbers are preferentially used for BBCs.)
APM30/TMC/ VIRTUAL/BBC/OM B	20 to 29	Remote cabinets used for monitoring GSM RRUs, such as power supply cabinets
Reserved	30 to 62	Others

Table 3-36 Cabinet configuration principles when indoor USU3910 modules are used

Logical Name of the Cabinet	Cabinet Number	Remarks
IBC10	0 to 7	Cabinets where the USU3910 resides, numbered according to the deployment sequence

3.1.3 Subrack

3.1.3.1 Subrack Types

Table 3-37 lists the subrack types.

Table 3-37 Subrack types

Logical Name of the Subrack	Remarks
BBU3900 (baseband unit)	BBU subrack: 2 U (88.9 mm or 3.50 in.)
BBU3910 (baseband unit)	BBU subrack: 2 U (88.9 mm or 3.50 in.)
BBU5900 (baseband unit)	BBU subrack: 2 U (88.9 mm or 3.50 in.)
BBU5910 (baseband unit)	BBU subrack: 2 U (88.9 mm or 3.50 in.)
BLADEBBU (baseband unit)	Outdoor BBU (BBU3910A) subrack
BBU3910C (baseband unit)	Outdoor BBU (BBU3910C) subrack
BBU5900A (baseband unit)	Outdoor BBU (BBU5900A) subrack
BOOKBBU5901 (baseband unit)	Outdoor BBU (BookBBU5901/ BookBBU5901a) subrack
BBU5900C (baseband unit)	Outdoor BBU (BBU5900C) subrack
BBU5900R (baseband Unit)	Outdoor integrated BTS (BTS5929R)
RFU (radio frequency unit)	RFU subrack
PMU (power monitoring unit)	Power supply subrack
TCU (temperature control unit)	Temperature control device subrack
FMU (fan monitoring unit)	Fan subrack
BAT (battery)	Battery subrack
RRU (radio remote unit)	RRU subrack
EMU (environment monitoring unit)	EMU subrack
GRU (GNSS receiving unit)	GRU subrack
CCU (cabinet control unit)	CCU subrack
EPU (embedded power unit)	EPU subrack
MEB (monitoring extension box)	MEB subrack
GDU (GNSS signal distribution unit)	GDU subrack
USU3910 (universal switching unit)	USU subrack: 2 U (88.9 mm or 3.50 in.)

Logical Name of the Subrack	Remarks
AAS (active antenna system)	AAS subrack
GATM	GATM subrack

3.1.3.2 Subrack Configuration Principles

Subracks can be entity subracks or extension subracks. Entity subracks, such as BBU3900 and RFU subracks, are fixed depending on the cabinet type. Extension subracks house optional peripherals, such as EMUs and GRUs. To ensure subrack number consistency and scalability, a peripheral can occupy an extension subrack independently.

Subrack numbers range from 0 to 254. Numbers of entity subracks cannot be changed. Numbers of extension subracks can be defined by users. [Table 3-38](#) to [Table 3-40](#) describe the configuration principles.

Table 3-38 Rules for numbering subracks for BTS3900/BTS5900 series base stations (including 3900 series upgraded ones)

Logical Name of the Subrack	Subrack Number	Subrack Type	Remarks
BBU3900/ BBU3910/ BLADEBBU/ BBU5900/BBU5910	0 or 1	Entity subrack	None
BBU5900A	0	Entity subrack	It can be configured only in the OBU5900A cabinet.
BOOKBBU5901	0	Entity subrack	It can be configured only in the OBU5901 cabinet.
BBU5900C	0	Entity subrack	It can be configured only in the OBU5900C cabinet.
BBU5900R	0	Virtual subrack	It can be configured only in the BTS5900R cabinet.
RFU	4 or 5	Entity subrack	RFU subrack. RFUs can be configured in subracks 4 and 5 in BTS3900L and virtual cabinets, and only in subrack 4 in other cabinets.
PMU	7	Entity subrack	The subrack where the OPM50M resides is also called PMU.
TCU	8	Entity subrack	None

Logical Name of the Subrack	Subrack Number	Subrack Type	Remarks
FMU	11 or 12	Entity subrack	None
CCU	15	Entity subrack	None
EPU	17 or 18	Entity subrack	None
GRU/EMU/ GATM/MEB/GDU	40 to 59	Extension subrack	You are advised to configure the GATM in subrack 50 or 51.
RRU	60 to 254	Entity subrack	See the recommended rules for numbering RRU subracks ^(a) .
AAS	60 to 254	Entity subrack	AAS subrack
Reserved	2 or 3	Reserved	None
Reserved	6	Reserved	None
Reserved	9 or 10	Reserved	None
Reserved	13 or 14	Reserved	The TCU can be configured in subrack 14 in a virtual cabinet.
Reserved	16, 19 to 39	Reserved	None

NOTE

(a) Recommended rules for numbering RRU subracks:

- The RRU subrack numbering rules for the same project must be consistent with each other.
- If a multimode RRU needs to be configured in multiple RATs, it must be configured with the same cabinet, subrack, and slot numbers. Otherwise an alarm is generated, indicating the configuration conflict.
- The RRU subrack numbers are numbered starting from the lowest band that the RRUs support. Some subrack numbers need to be reserved for capacity expansion. For example, in a network using RRU3929s in 1800 MHz and RRU3806s in 2100 MHz, the subrack numbers are assigned as follows: 70 to 79 are assigned to the RRU3929s in 1800 MHz, and 80 to 89 are assigned to the RRU3806s in 2100 MHz.

Table 3-39 Rules for numbering subracks for upgraded BTS3012 series base stations

Logical Name of the Subrack	Subrack Number	Subrack Type	Remarks
BBU3900/BBU3910	0	Entity subrack	None
BBU3900/BBU3910	1	Entity subrack	None
RFU	4	Entity subrack	None
RFU	5	Entity subrack	RFU subrack, which does not house any board
PMU	7	Entity subrack	None
TCU	8	Entity subrack	None
FMU	11	Entity subrack	None
RRU	60 to 254	Entity subrack	RRU subrack, which must be configured in cabinet 0
Reserved	2 or 3	Reserved	None
Reserved	6	Reserved	None
Reserved	9 or 10	Reserved	None
Reserved	12 to 19	Reserved	None
MCS	20	Entity subrack	Common subrack, which does not house any board
AFS	21	Entity subrack	Combiner subrack, which does not house any board
Reserved	22 to 39	Reserved	None
Reserved	40 to 59	Extension subrack	For housing the EMU and other boards

Table 3-40 Rules for numbering USU3910 subracks

Logical Name of the Subrack	Subrack Number	Subrack Type	Remarks
USU3910	0 or 1	Entity subrack	None

3.1.4 BBU Board Types

BBUs include the BBU3900/BBU3910/BBU5900/BBU5910/BBU3910C/BBU3910A/BBU5900A/BBU5900C/BookBBU5901/BookBBU5901a.

Table 3-41 lists the boards that can be configured in a BBU3900 and the corresponding configuration rules.

Table 3-41 Configurations of boards in a BBU3900

Board Name	Logic al Name	Cabin et Number	Subra ck Number	Slot Number	Quantity
LTE baseband processing unit (LBBP)	LBBP	0 to 7	0 or 1	0 to 5	0 to 6
Universal transmission processing unit (UTRP)	UTRP	0 to 7	0 or 1	UMTS/LTE: 0 to 6	0 or 1
Universal satellite card and clock unit (USCU)	USCU	0 to 7	0 or 1	0 to 5	0 or 1
GSM transmission and timing and management unit (GTMU)	GTMU	0 to 7	0 or 1	6	0 or 1
Universal main processing and transmission unit (UMPT)	UMPT	0 to 7	0 or 1	6 or 7	0 to 2
Fan module of the BBU (FAN)	FAN	0 to 7	0 or 1	16	1
Universal power and environment interface unit (UPEU)	UPEU	0 to 7	0 or 1	18 or 19	1 or 2
Universal environment interface unit (UEIU)	UEIU	0 to 7	0 or 1	18 or 19	0 or 1
Universal inter-connection infrastructure unit (UCIU)	UCIU	0 to 7	0 or 1	0 to 7	0 or 1
Universal baseband radio interface board (UBRI)	UBRI	0 to 7	0 or 1	0 to 3	0 to 2
Universal baseband processing unit (UBBP)	UBBP	0 to 7	0 or 1	0 to 5	0 to 6

Table 3-42 lists the boards that can be configured in a BBU3910 and the corresponding configuration rules.

Table 3-42 Configurations of boards in a BBU3910

Board Name	Logic al Nam e	Cabin et Num ber	Subra ck Num ber	Slot Number	Quantity
Universal transmission processing unit (UTRP)	UTRP	0 to 7	0 or 1	0 to 6	0 or 1
Universal satellite card and clock unit (USCU)	USCU	0 to 7	0 or 1	0 to 5	0 or 1
Universal main processing and transmission unit (UMPT)	UMPT	0 to 7	0 or 1	6 or 7	0 to 2
GSM transmission and timing and management unit (GTMUc)	GTMU	0 to 7	0 or 1	6	0 or 1
Fan module of the BBU (FAN)	FAN	0 to 7	0 or 1	16	1
Universal power and environment interface unit (UPEU)	UPEU	0 to 7	0 or 1	18 or 19	1 or 2
Universal environment interface unit (UEIU)	UEIU	0 to 7	0 or 1	18 or 19	0 or 1
Universal baseband radio interface board (UBRI)	UBRI	0 to 7	0 or 1	0 to 5	0 to 2
Universal baseband processing unit (UBBP)	UBBP	0 to 7	0 or 1	0 to 5	0 to 6

Table 3-43 lists the boards that can be configured in a BBU5900 and the corresponding configuration rules.

Table 3-43 Configurations of boards in a BBU5900

Board Name	Logic al Nam e	Cabin et Num ber	Subra ck Num ber	Slot Number	Quantity
Universal satellite card and clock unit (USCU)	USCU	0 to 7	0 or 1	0 to 5	0 or 1
Universal main processing and transmission unit (UMPT)	UMPT	0 to 7	0 or 1	6 or 7	1 or 2

Board Name	Logical Name	Cabinet Number	Subrack Number	Slot Number	Quantity
Fan module of the BBU (FAN)	FAN	0 to 7	0 or 1	16	1
Universal power and environment interface unit (UPEU)	UPEU	0 to 7	0 or 1	18 or 19	1 or 2
Universal environment interface unit (UEIU)	UEIU	0 to 7	0 or 1	18 or 19	0 or 1
Universal baseband processing unit (UBBP)	UBBP	0 to 7	0 or 1	UBBPfw: 0, 2, or 4 Other boards: 0 to 5	UBBPfw: 0 to 3 Other boards: 0 to 6

Table 3-44 lists the boards that can be configured in a BBU5910 and the corresponding configuration rules.

Table 3-44 Configurations of boards in a BBU5910

Board Name	Logical Name	Cabinet Number	Subrack Number	Slot Number	Quantity
Universal satellite card and clock unit (USCU)	USCU	0 to 7	0 or 1	0 to 5	0 or 1
Universal main processing and transmission unit (UMPT)	UMPT	0 to 7	0 or 1	6 or 7	1 or 2
Fan module of the BBU (FAN)	FAN	0 to 7	0 or 1	16	1
Universal power and environment interface unit (UPEU)	UPEU	0 to 7	0 or 1	18 or 19	1 or 2
Universal environment interface unit (UEIU)	UEIU	0 to 7	0 or 1	18 or 19	0 or 1
Universal baseband processing unit (UBBP)	UBBP	0 to 7	0 or 1	0 to 5	0 to 6

 NOTE

- BBU boards in a GBTS support automatic or manual configuration. If a BBU board supports automatic configuration, the configuration of the board automatically takes effect when the configuration of the cabinet housing the board takes effect. If a BBU board supports manual configuration, the configuration of the board takes effect only after the board is manually configured. In a GBTS BBU, the FAN, GTMU, and UPEU support automatic configuration, whereas other boards require manual configuration.
- All boards installed in a BBU must be configured.
- During BBU3900/BBU3910 configuration, the number of UMPT boards can be 0 only when the GTMU is present in the BBU3900/BBU3910.

Table 3-45 lists the boards that can be configured in a BBU3910A and the corresponding configuration rules.

Table 3-45 Configurations of boards in a BBU3910A

Board Name	Logical Name	Cabinet Number	Subrack Number	Slot Number	Quantity
Universal multimode digital unit (UMDU)	UMDU	0 to 7	0 or 1	6	1

Table 3-46 lists the boards that can be configured in a BBU3910C and the corresponding configuration rules.

Table 3-46 Configurations of boards in a BBU3910C

Board Name	Logical Name	Cabinet Number	Subrack Number	Slot Number	Quantity
Multimode digital unit (compact) (MDUC)	MDUC	0 to 7	0	6	1

Table 3-47 lists the boards that can be configured in a BBU5900A and the corresponding configuration rules.

Table 3-47 Configurations of boards in a BBU5900A

Board Name	Logic al Nam e	Cabin et Num ber	Subra ck Num ber	Slot Number	Quantity
Universal main processing and transmission unit (UMPT)	UMPT	0 to 7	0	6	1
Fan module of the BBU (FAN)	FAN	0 to 7	0	16 or 17	2
Universal power and environment interface unit (UPEU)	UPEU	0 to 7	0	19	1
Universal environment interface unit (UEIU)	UEIU	0 to 7	0	18	1
Universal baseband processing unit (UBBP)	UBBP	0 to 7	0	0, 2, or 4	1 to 3

Table 3-48 lists the boards that can be configured in a BookBBU5901 and the corresponding configuration rules.

Table 3-48 Configurations of boards in a BookBBU5901

Board Name	Logic al Nam e	Cabin et Num ber	Subra ck Num ber	Slot Number	Quantity
Universal main processing digital unit (UMDU)	UMDU	0 to 7	0	6	1

Table 3-49 lists the boards that can be configured in a BBU5900C and the corresponding configuration rules.

Table 3-49 Configurations of boards in a BBU5900C

Board Name	Logic al Nam e	Cabin et Num ber	Subra ck Num ber	Slot Number	Quantity
Universal main processing digital unit (UMDU)	UMDU	0 to 7	0	6	1

Table 3-50 lists the boards that can be configured in a BookBBU5901a and the corresponding configuration rules.

Table 3-50 Configurations of boards in a BookBBU5901a

Board Name	Logical Name	Cabinet Number	Subrack Number	Slot Number	Quantity
Universal main processing digital unit (UMDU)	UMDU	0 to 7	0	6	1

Table 3-51 lists the boards that can be configured in a BTS5929R and the corresponding configuration rules.

Table 3-51 Configurations of boards in a BTS5929R

Board Name	Logical Name	Cabinet Number	Subrack Number	Slot Number	Quantity
Universal main processing digital unit (UMDU)	UMDU	0 to 7	0	6	1

 NOTE

All boards installed in a BBU must be configured.

The BBU slot assignment in this section involves only the software configuration of the slot number for each board. The slot in which a board is installed is determined by the RAT. For details, see hardware descriptions of the corresponding products in *3900 & 5900 Series Base Station Product Documentation*.

3.1.5 USU Board Types

Table 3-52 lists the boards that can be configured in a USU3910 and their configuration rules.

Table 3-52 Configurations of boards in a USU3910

Board Name	Logical Name	Cabinet Number	Subrack Number	Slot Number	Quantity
Universal enhanced switch fabric unit (UEFU)	UEFU	0 to 7	0 or 1	0 or 1	2

Board Name	Logical Name	Cabinet Number	Subrack Number	Slot Number	Quantity
Universal line process unit (ULPU)	ULPU	0 to 7	0 or 1	1	1
Universal power input unit (UPIU)	UPIU	0 to 7	0 or 1	18 or 19	2
Universal fan control unit (UFCU)	UFCU	0 to 7	0 or 1	16 or 17	2

3.1.6 Typical Cabinet and Subrack Configurations for 5900 Series Base Stations

The configuration rules in this section apply to eGBTs, NodeBs, eNodeBs, gNodeBs, and co-MPT base stations.

In multimode base stations, the configuration rules of cabinets, subracks, and common boards in BBU subracks are listed as follows:

- The primary RAT can be one of the following RATs with a priority in descending order: GSM > UMTS > LTE > NR.
- For the primary RAT, all the following must be configured: cabinets, BBU subracks, RF subracks used by the primary RAT, PMU subracks, EMU subracks, TCU subracks, CCU subracks, and FMU subracks. However, configuration for the secondary RAT involves only the following: BBU subracks, RF subracks used by the secondary RAT, and the cabinets housing these subracks.
- The configuration of a cabinet number or subrack number must keep consistent if the cabinet or subrack is configured in both the primary RAT and secondary RAT. Otherwise, configuration conflict alarms will be reported.
- In a BBU subrack, the UPEU, UEIU, and FAN can be configured in two modes. The USCUC can be configured in either one mode or two modes.

3.1.6.1 BTS5900 Base Stations

The following tables list the typical cabinet and subrack configurations for BTS5900 base stations.

Scenario 1: -48 V DC Power Input

Table 3-53 Typical cabinet and subrack configurations for BTS5900 base stations (-48 V DC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BTS5900	1	BTS5900	0 or 1	RFU	4	None
				FMU	11	
				EMU	40	
				BBU5900/BBU5910	0	
BTS3900	1	BTS3900	0 or 1	RFU	4	
				FMU	11	
				EMU	40	
				BBU5900/BBU5910	0	

Scenario 2: +24 V DC, 220 V AC, or 110 V AC

Table 3-54 Typical cabinet and subrack configurations for BTS5900 base stations (+24 V DC, 220 V AC, or 110 V AC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BTS3900	1 or 2	BTS3900	0 or 1	RFU	4	If a base station is configured with two cabinets, it is recommended that the cabinets be numbered 0 and 1, respectively. The BBU is configured in cabinet 0 and manages both cabinets. Each cabinet is configured with one PMU. A +24 V cabinet does not require any PMU.
				FMU	11	
				BBU5900/BBU5910	0	
				EMU	40	
				PMU	7	

3.1.6.2 BTS5900L Base Stations

The following tables list the typical cabinet and subrack configurations for BTS5900L base stations.

Scenario 1: -48 V DC Power Input

Table 3-55 Typical cabinet and subrack configurations for BTS5900L base stations (-48 V DC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BTS5900L	1	BTS5900L	0	RFU	4 or 5	For a BTS3900L cabinet, the lower and upper RFU subrack numbers are 4 and 5, respectively. The lower and upper FMU subrack numbers are 11 and 12, respectively.
				FMU	11 or 12	
				EMU	40	
				BBU5900/BBU5910	0	
BTS3900L	1	BTS3900L	0	RFU	4 or 5	
				FMU	11 or 12	
				EMU	40	
				BBU5900/BBU5910	0	

Scenario 2: 220 V AC or 110 V AC

Table 3-56 Typical cabinet and subrack configurations for BTS5900L base stations (220 V AC or 110 V AC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BTS3900L	1	BTS3900L	0	RFU	4 or 5	For a BTS3900L cabinet, the lower and upper RFU subrack numbers are 4 and 5, respectively. The lower and upper FMU subrack numbers are 11 and 12, respectively.
				FMU	11 or 12	
				EMU	40	
				PMU	7	
				BBU5900/BBU5910	0	

3.1.6.3 BTS5900A Base Stations

The following tables list the typical cabinet and subrack configurations for BTS5900A base stations.

Scenario 1: -48 V DC Power Input

Table 3-57 Typical cabinet and subrack configurations for BTS5900A base stations (-48 V DC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
TMC11H	1	TMC	0	TCU	8	None
				BBU5900/BBU5910	0	None

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
				CCU	15	This subrack is configured only when intelligent features are required for Ver.E cabinets.
RFC	1 to 4	RFC	1 to 4	RFU	4	None
				FMU	11	None
RFC5906	0 to 4	RFC5906	1 to 4	FMU	11	None

Scenario 2: 220 V AC Power Input

Table 3-58 Typical cabinet and subrack configurations for BTS5900A base stations (220 V AC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
APM30H	1 to 3	APM30	0, 5, or 6	TCU	8	APM30 cabinets connect to the monitoring devices in multiple cabinets.
				BBU5900/BBU5910	0	None
				PMU	7	None
				EMU	40	These subracks are configured in No.0 APM30.
				GATM0	50	None
				GATM1	51	None

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
				CCU	15	This subrack is configured only when intelligent features are required for Ver.E cabinets.
RFC	1 to 4	RFC	1 to 4	RFU	4	None
				FMU	11	
RFC5906	0 to 4	RFC5906	1 to 4	FMU	11	None
TMC11	0 or 1	TMC	7 or 8 ^(a)	TCU	8	None
				CCU	15	This subrack is configured only when intelligent features are required for Ver.E cabinets.
IBBS200D, IBBS200T, IBBS300D, IBBS300T, or BBC5200D-L	0 to 6	BBC	9 to 14	TCU	8	APM30 cabinets connect to the monitoring devices in BBCs. - No.0 APM30 connects to No.9 and No.10 BBCs. - No.5 APM30 connects to No.11 and No.12 BBCs. - No.6 APM30 connects to No.13 and No.14 BBCs.

 NOTE

(a) It is recommended that a transmission cabinet configured with a BBU be numbered 7. If conditions do not allow, configure the transmission cabinet as cabinet 8.

3.1.6.4 DBS5900 Base Stations

The following tables list the typical cabinet and subrack configurations for DBS5900 base stations.

Scenario 1: -48 V DC Power Input

Table 3-59 Typical cabinet and subrack configurations for DBS5900 base stations (-48 V DC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
APM5930(DC)	1	APM5930	0	TCU	8	In a triple-mode co-cabinet scenario, two APM5930(DC) cabinets must be configured and each cabinet is equipped with one BBU.
				BBU5900/BBU5910	0	
APM5900H or APM5900H-L	1	APM5900	0	TCU	8	None
				BBU5900/BBU5910	0	
IBC10(DC)	0 or 1	IBC10	0	BBU5900/BBU5910	0	None
IFS5906	0 to 4	IFS5906	1 to 4	FMU	11	The FMU board is configured only when the fan assembly is configured.
RFC5906	0 to 4	RFC5906	1 to 4	FMU	11	None

Scenario 2: 110 V AC or 220 V AC Power Input

Table 3-60 Typical cabinet and subrack configurations for DBS5900 base stations (110 V AC or 220 V AC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
VIRTUAL	0 or 1	VIRTUAL	0	BBU5900/BBU5910	0	None
APM5930(AC)	1 to 3	APM5930	0, 5, or 6	TCU	8	APM5930 cabinets manage monitoring devices in multiple cabinets.
				BBU5900/BBU5910	0	None
				PMU	7	None
APM5930(DC)	0 or 1	APM5930	7 or 8(a)	TCU	8	None
APM5950H or APM5950H-L	1 to 3	APM5950	0, 5, or 6	BBU5900/BBU5910	0	None
				PMU	7	None
OPM200	0 to 2	OPM	5 or 6	PMU	7	None
IFS5906	0 to 4	IFS5906	1 to 4	FMU	11	The FMU board is configured only when the fan assembly is configured.
RFC5906	0 to 4	RFC5906	1 to 4	FMU	11	None

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BBC520 OD, BBC520 OT, BBC530 OD, BBC530 OT, or BBC520 OD-L	0 to 6	BBC	9 to 14	TCU	8	APM5930 cabinets connect to the monitoring devices in BBCs. - No.0 APM5930 connects to No.9 and No.10 BBCs. - No.5 APM5930 connects to No.11 and No.12 BBCs. - No.6 APM5930 connects to No.13 and No.14 BBCs.

 NOTE

(a) It is recommended that a transmission cabinet configured with a BBU be numbered 7. If conditions do not allow, configure the transmission cabinet as cabinet 8.

3.1.6.5 DBS5900E Base Stations

The following tables list the typical cabinet and subrack configurations for DBS5900E base stations.

Table 3-61 Typical cabinet and subrack configurations for DBS5900E base stations (outdoor BookBBU5901 with 110 V AC or 220 V AC power input)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
OBU5901	1	OBU5901	0	BookBBU5901	0	None

Table 3-62 Typical cabinet and subrack configurations for DBS5900E base stations (outdoor BBU5900C with -48 V DC power input)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
OBU5900C	1	OBU5900C	0	BBU5900C	0	None

3.1.6.6 BTS5900R Base Station

The following tables list the typical cabinet and subrack configurations for BTS5900R base stations.

Table 3-63 Typical cabinet and subrack configurations for BTS5900R base stations (outdoor BTS5929R with -48 V DC power input)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BTS5929R	1	BTS5900R	0	BBU5900R	0	None

3.1.6.7 USUs

The following table lists the typical cabinet and subrack configurations for USUs.

Table 3-64 Typical cabinet and subrack configurations for USU3910s (-48 V DC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
Virtual	1	VIRTUAL	0	USU3910	0	None

3.1.7 Typical Cabinet and Subrack Configurations for 3900 Series Base Stations

The configuration rules in this section apply to GBTs, eGBTs, NodeBs, eNodeBs, gNodeBs, and co-MPT base stations.

In multimode base stations, the configuration rules of cabinets, subracks, and common boards in BBU subracks are listed as follows:

- The primary RAT can be one of the following RATs with a priority in descending order: GSM > UMTS > LTE > NR.
- For the primary RAT, all the following must be configured: cabinets, BBU subracks, RF subracks used by the primary RAT, PMU subracks, EMU subracks, TCU subracks, CCU subracks, and FMU subracks. However, configuration for the secondary RAT involves only the following: BBU subracks, RF subracks used by the secondary RAT, and the cabinets housing these subracks.
- The configuration of a cabinet number or subrack number must keep consistent if the cabinet or subrack is configured in both the primary RAT and secondary RAT. Otherwise, configuration conflict alarms will be reported.
- In a BBU subrack, the UPEU, UEIU, and FAN can be configured in two RATs. The USCU can be configured in either one RAT or two RATs, and the UCIU and UTRPc can be configured only in one RAT.

3.1.7.1 BTS3900 Base Stations

The following tables list the typical cabinet and subrack configurations for BTS3900 base stations.

Scenario 1: -48 V DC Power Input

Table 3-65 Typical cabinet and subrack configurations for BTS3900 base stations (-48 V DC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BTS3900	1	BTS3900	0 or 1	RFU	4	None
				FMU	11	
				EMU	40	
				BBU3900 or BBU3910	0	
BTS3012_SRAN	1 or 2	BTS3012_SRAN	0 or 1	RFU	4	If a base station is configured with two cabinets, it is recommended that the cabinets be numbered 0 and 1, respectively. The BBU is configured in cabinet 0 and manages both cabinets.
				FMU	11	
				BBU3900 or BBU3910	0	
				EMU	40	

Scenario 2: 220 V AC Power Input

Table 3-66 Typical cabinet and subrack configurations for BTS3900 base stations (220 V AC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BTS3900	1 or 2	BTS3900	0 or 1	RFU	4	- If a base station is configured with two cabinets, it is recommended that the cabinets be numbered 0 and 1, respectively. The BBU is configured in cabinet 0 and manages both cabinets. - Each cabinet is configured with one PMU.
				FMU	11	
				EMU	40	
				BBU3900 or BBU3910	0	
				PMU	7	
BTS3012_SRAN	1 or 2	BTS3012_SRAN	0 or 1	RFU	4	If a base station is configured with two cabinets, it is recommended that the cabinets be numbered 0 and 1, respectively. The BBU is configured in cabinet 0 and manages both cabinets.
				FMU	11	
				BBU3900 or BBU3910	0	
				EMU	40	
				PMU	7	

Scenario 3: +24 V DC or 110 V AC

Table 3-67 Typical cabinet and subrack configurations for BTS3900 base stations (+24 V DC or 110 V AC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BTS3900	1 or 2	BTS3900	0 or 1	RFU	4	- If a base station is configured with two cabinets, it is recommended that the cabinets be numbered 0 and 1, respectively. The BBU is configured in cabinet 0 and manages both cabinets. - Each cabinet is configured with one PMU. A +24 V cabinet does not require any PMU.
				FMU	11	
				BBU3900 or BBU3910	0	
				EMU	40	
				PMU	7	

3.1.7.2 BTS3900L Base Stations

The following tables list the typical cabinet and subrack configurations for BTS3900L base stations.

Scenario 1: -48 V DC Power Input

Table 3-68 Typical cabinet and subrack configurations for BTS3900L base stations (-48 V DC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BTS3900L	1	BTS3900L	0	RFU	4 or 5	For a BTS3900L cabinet, the lower and upper RFU subrack numbers are 4 and 5, respectively. The lower and upper FMU subrack numbers are 11 and 12, respectively.
				FMU	11 or 12	
				EMU	40	
				GATM0	50	
				GATM1	51	
				BBU3900 or BBU3910	0	

Scenario 2: 220 V AC or 110 V AC

Table 3-69 Typical cabinet and subrack configurations for BTS3900L base stations (220 V AC or 110 V AC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BTS3900L	1	BTS3900L	0	RFU	4 or 5	For a BTS3900L cabinet, the lower and upper RFU subrack numbers are 4 and 5, respectively. The lower and upper FMU subrack numbers are 11 and 12, respectively.
				FMU	11 or 12	
				EMU	40	
				PMU	7	
				GATM0	50	
				GATM1	51	
				BBU3900 or BBU3910	0	

3.1.7.3 BTS3900A Base Stations

The following tables list the typical cabinet and subrack configurations for BTS3900A base stations.

Scenario 1: -48 V DC Power Input

Table 3-70 Typical cabinet and subrack configurations for BTS3900A base stations (-48 V DC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
TMC11H	1	TMC	0	TCU	8	None
				BBU3900 or BBU3910	0	
				CCU	15	This subrack is configured only when intelligent features are required for Ver.E cabinets.
RFC	1 to 4	RFC	1 to 4	RFU	4	None
				FMU	11	
RFC5906	0 to 4	RFC5906	1 to 4	FMU	11	None

Scenario 2: 220 V AC Power Input

Table 3-71 Typical cabinet and subrack configurations for BTS3900A base stations (220 V AC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
APM30H	1 to 3	APM30	0, 5, or 6	TCU	8	APM30 cabinets connect to the monitoring devices in multiple cabinets.
				BBU3900 or BBU3910	0	

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
				PMU	7	These subracks are configured in No.0 APM30.
				EMU	40	
				GATM0	50	
				GATM1	51	
				CCU	15	This subrack is configured only when intelligent features are required for Ver.E cabinets.
RFC	1 to 4	RFC	1 to 4	RFU	4	None
				FMU	11	
RFC5906	0 to 4	RFC5906	1 to 4	FMU	11	None
TMC11	0 or 1	TMC	7 or 8 ^(a)	TCU	8	None
				CCU	15	
IBBS200D, IBBS200T, IBBS300D, IBBS300T, or BBC5200D-L	0 to 6	BBC	9 to 14	TCU	8	APM30 cabinets connect to the monitoring devices in BBCs. - No.0 APM30 connects to No.9 and No.10 BBCs. - No.5 APM30 connects to No.11 and No.12 BBCs. - No.6 APM30 connects to No.13 and No.14 BBCs.
BTS3012AE_SRAN	1 or 2	BTS3012AE_SRAN	0 or 1	TCU	8	None
				RFU	4	

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
				BBU3900 or BBU3910	0	
				PMU	7	
				FMU	11	
				EMU	40	
IBBS200D, IBBS200T, IBBS700D, or IBBS700T	0 to 4	BBC	9 to 12	TCU	8	BTS3012AE_SRRN cabinets connect to the monitoring devices in BBCs. - No.0 cabinet manages No.9 and No.10 BBCs when requiring two battery cabinets. - No.1 cabinet manages No.11 and No.12 BBCs when requiring two battery cabinets.

 NOTE

(a) It is recommended that a transmission cabinet configured with a BBU be numbered 7. If conditions do not allow, configure the transmission cabinet as cabinet 8.

3.1.7.4 BTS3900AL Base Stations

The following table lists the typical cabinet and subrack configurations for BTS3900AL base stations.

Table 3-72 Typical cabinet and subrack configurations for BTS3900AL base stations (220 V AC)

Hardw are Cabine t Type	Quant ity of the Cabin et to Be Confi gured	Name of the Cabinet to Be Configure d	Cab inet Nu mb er	Recomme nded Subrack Type	Sub rac k Nu mb er	Remarks
BTS3900AL	1 or 2	BTS3900AL	0 or 1	BBU3900 or BBU3910	0	None
				BBU3900 or BBU3910	1	
				PMU	7	
				TCU	8	
				CCU	15	
				FMU	11	
				EMU	40	
				RFU	4	
TMC11H	0 or 1	TMC	8	TCU	8	
IBBS700D, IBBS700T, IBBS300D, or IBBS300T	1 or 2	BBC	9 or 10	CCU	15	
				TCU	8	

3.1.7.5 DBS3900 Base Stations

The following tables list the typical cabinet and subrack configurations for DBS3900 base stations.

Scenario 1: -48 V DC Power Input (Used with a BBU3900 or BBU3910)

Table 3-73 Typical cabinet and subrack configurations for DBS3900 base stations (BBU3900 or BBU3910, -48 V DC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
TMC or TMC11H	0 or 1	TMC	0	TCU	8	In a triple-mode co-cabinet scenario, two TMCs must be configured and each cabinet is equipped with one BBU.
				BBU3900 or BBU3910	0	
				CCU	15	This subrack is configured only when intelligent features are required for Ver.E cabinets.
OMB	0 or 1	OMB	0	TCU	8	None
				BBU3900 or BBU3910	0	
IBC10 (DC)	0 or 1	IBC10	0	BBU3900 or BBU3910	0	None
RFC5906	0 to 4	RFC5906	1 to 4	FMU	11	None

Scenario 2: +24 V DC Power Input (Used with a BBU3900 or BBU3910)

Table 3-74 Typical cabinet and subrack configurations for DBS3900 base stations (BBU3900 or BBU3910, +24 V DC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
APM30 or APM30H	1 or 2	APM30	0	TCU	8	In a triple-mode co-cabinet scenario, two APM30 cabinets must be configured and each cabinet is equipped with one BBU.
				BBU3900 or BBU3910	0	
				CCU	15	This subrack is configured only when intelligent features are required for Ver.E cabinets.
APM5930(DC)	1	APM5930	0	TCU	8	In a triple-mode co-cabinet scenario, two APM5930(DC) cabinets must be configured and each cabinet is equipped with one BBU.
				BBU3900 or BBU3910	0	
APM5900H/ APM5900H-L	1	APM5900	0	TCU	8	None
				BBU3900 or BBU3910	0	
RFC5906	0 to 4	RFC5906	1 to 4	FMU	11	None

Scenario 3: 220 V AC Power Input (Used with a BBU3900 or BBU3910)

Table 3-75 Typical cabinet and subrack configurations for DBS3900 base stations (BBU3900 or BBU3910, 220 V AC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
VIRTUAL	0 or 1	VIRTUAL	0	BBU3900 or BBU3910	0	None
APM30 or APM30H	1 to 3	APM30	0, 5, or 6	TCU	8	APM30 cabinets manage monitoring devices in multiple cabinets.
				BBU3900 or BBU3910	0	
				PMU	7	
				CCU	15	This subrack is configured only when intelligent features are required for Ver.E cabinets.
APM5930(AC)	1 to 3	APM5930	0, 5, or 6	TCU	8	APM5930 cabinets manage monitoring devices in multiple cabinets.
				BBU3900 or BBU3910	0	None
				PMU	7	None
APM5930(DC)	0 or 1	APM5930	7 or 8(a)	TCU	8	None
APM5950H or APM5950H-L	1 to 3	APM5950	0, 5, or 6	BBU3900 or BBU3910	0	None
				PMU	7	None
APM5900H or APM5900H-L	0 or 1	APM5900	7 or 8(a)	TCU	8	None

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
TMC or TMC11H	0 or 1	TMC	7 or 8(a)	TCU	8	None
				CCU	15	This subrack is configured only when intelligent features are required for Ver.E cabinets.
OMB	0 or 1	OMB	0	TCU	8	When the OMB is configured, other cabinets are not required.
				BBU3900 or BBU3910	0	
				PMU	7	
IBBS20OD, IBBS20OT, IBBS30OD, IBBS30OT, IBBS20D, or BBC520OD-L	0 to 6	BBC	9 to 14	TCU	8	OMB cabinets connect to the monitoring devices in IBBS20D cabinets. APM30 cabinets connect to the monitoring devices in BBCs other than IBBS20D. - No.0 APM30 or No.0 OMB connects to No.9 and No.10 BBCs. - No.5 APM30 or No.1 OMB connects to No.11 and No.12 BBCs. - No.6 APM30 connects to No.13 and No.14 BBCs.
TP4860OA	0 or 1	TP48600A	0	PMU	7	None
				CCU	15	
				TCU	8	
				EMU	40	

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
				BBU3900 or BBU3910	0	
RFC5906	0 to 4	RFC5906	1 to 4	FMU	11	None
IFS5906	0 to 4	IFS5906	1 to 4	FMU	11	The FMU board is configured only when the fan assembly is configured.
OPM200	0 to 2	OPM	5 or 6	PMU	7	None

 NOTE

(a) It is recommended that a transmission cabinet configured with a BBU be numbered 7. If conditions do not allow, configure the transmission cabinet as cabinet 8.

Scenario 4: 110 V AC Power Input (Used with a BBU3900 or BBU3910)

Table 3-76 Typical cabinet and subrack configurations for DBS3900 base stations (BBU3900 or BBU3910, 110 V AC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
VIRTUAL	0 or 1	VIRTUAL	0	BBU3900 or BBU3910	0	None
APM30 or APM30H	1 to 3	APM30	0, 5, or 6	TCU	8	APM30 cabinets manage monitoring devices in multiple cabinets.
				BBU3900 or BBU3910	0	

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
				PMU	7	
				CCU	15	
				This subrack is configured only when intelligent features are required for Ver.E cabinets.		
APM5930(AC)	1 to 3	APM5930	0, 5, or 6	TCU	8	APM5930 cabinets manage monitoring devices in multiple cabinets.
				BBU3900 or BBU3910	0	None
				PMU	7	None
APM5930(DC)	0 or 1	APM5930	7 or 8 ^(a)	TCU	8	None
APM5950H or APM5950H-L	1 to 3	APM5950	0, 5, or 6	BBU3900 or BBU3910	0	None
				PMU	7	None
APM5900H or APM5900H-L	0 or 1	APM5900	7 or 8 ^(a)	TCU	8	None
TMC or TMC11H	0 or 1	TMC	7 or 8 ^(a)	TCU	8	None
				CCU	15	This subrack is configured only when intelligent features are required for Ver.E cabinets.
OMB	0 or 1	OMB	0	TCU	8	When the OMB is configured, other cabinets are not required.
				BBU3900 or BBU3910	0	

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
				PMU	7	
IBBS200D, IBBS200T, IBBS300D, IBBS300T, IBBS200D, or BBC5200D-L	0 to 6	BBC	9 to 14	TCU	8	OMB cabinets connect to the monitoring devices in IBBS20D cabinets. APM30 cabinets connect to the monitoring devices in BBCs other than IBBS20D. - No.0 APM30 or No.0 OMB connects to No.9 and No.10 BBCs. - No.5 APM30 or No.1 OMB connects to No.11 and No.12 BBCs. - No.6 APM30 connects to No.13 and No.14 BBCs.
TP48600A	0 or 1	TP48600A	0	PMU	7	None
				CCU	15	
				TCU	8	
				EMU	40	
				BBU3900 or BBU3910	0	
RFC5906	0 to 4	RFC5906	1 to 4	FMU	11	None
IFS5906	0 to 4	IFS5906	1 to 4	FMU	11	The FMU board is configured only when the fan assembly is configured.
OPM200	0 to 2	OPM	5 or 6	PMU	7	None

 NOTE

(a) It is recommended that a transmission cabinet configured with a BBU be numbered 7. If conditions do not allow, configure the transmission cabinet as cabinet 8.

Scenario 5: -48 V DC Power Input (Used with a BBU3910A)

Table 3-77 Typical cabinet and subrack configurations for DBS3900 base stations (BBU3910A, -48 V DC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BBU3910A	1	VIRTUAL	0	BLADEBBU	0	None

Scenario 6: 220 V AC Power Input (Used with a BBU3910A)

Table 3-78 Typical cabinet and subrack configurations for DBS3900 base stations (BBU3910A, 220 V AC, OPM50M)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BBU3910A	1	VIRTUAL	0	BLADEBBU	0	None
OPM50M	1 to 3	VIRTUAL	0, 5, or 6	PMU	7	None
OPM200	0 to 2	OPM	5 or 6	PMU	7	None
IBBS20D	0 to 6	BBC	9 to 14	TCU	8	None

Table 3-79 Typical cabinet and subrack configurations for DBS3900 base stations (BBU3910A, 220 V AC, OPM15M)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
BBU3910A	1	VIRTUAL	0	BLADEBBU	0	No configuration is required for an OPM15M.

Scenario 7: -48 V DC Power Input (Used with a BBU3910C)

Table 3-80 Typical cabinet and subrack configurations for DBS3900 base stations (BBU3910C, -48 V DC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
Virtual cabinet	1	VIRTUAL	0	BBU3910C	0	None

3.1.7.6 BTS3900C Base Stations

Scenario 1: -48 V DC Power Input

Table 3-81 Typical cabinet and subrack configurations for BTS3900C base stations (-48 V DC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
OMB	1	OMB	0	BBU3900/ BBU3910	0	None
				TCU	8	

Scenario 2: 110 V AC or 220 V AC Power Input

Table 3-82 Typical cabinet and subrack configurations for BTS3900C base stations (110 V AC or 220 V AC)

Hardware Cabinet Type	Quantity of the Cabinet to Be Configured	Name of the Cabinet to Be Configured	Cabinet Number	Recommended Subrack Type	Subrack Number	Remarks
OMB	1	OMB	0	TCU	8	None
				BBU3900/ BBU3910	0	
				PMU	7	

3.1.7.7 USUs

Table 3-83 lists the typical configurations of USUs.

Table 3-83 Typical configurations of USUs

Item	USU3910
Typical Installation Scenario	-48 V DC power input
Hardware Cabinet Type	VIRTUAL
Quantity of the Cabinet to Be Configured	1
Name of the Cabinet to Be Configured	VIRTUAL
Cabinet Number	0
Recommended Subrack Type	USU3910
Subrack Number	0

3.2 Network Analysis

3.2.1 Benefits

None

3.2.2 Impacts

The impacts between this function and other functions are described in [Function Impacts](#).

This function has no impact on the network.

3.3 Requirements

3.3.1 Licenses

None

3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.3.2.1 Prerequisite Functions

None

3.3.2.2 Mutually Exclusive Functions

None

3.3.2.3 Function Impacts

None

3.3.3 Hardware

3900 and 5900 series base stations (macro base stations)

3.3.4 Others

None

3.4 Operation and Maintenance (GBTS)

3.4.1 Data Configuration

3.4.1.1 Data Preparation

Scenario 1: Adding Cabinets

Table 3-84 describes the parameters for adding cabinets.

Table 3-84 Parameters for adding cabinets to GBTSs

Parameter Name	Parameter ID	Setting Notes
Index Type	BTSCABINET.IDTYPE	Set this parameter to BYNAME or BYID based on site requirements.
BTS Index	BTSCABINET.BTSD	This parameter uniquely identifies a base station under a BSC.
BTS Name	BTSCABINET.BTSNAME	This parameter uniquely identifies a base station under a BSC.
Cabinet No.	BTSCABINET.CN	It is recommended that the cabinet number start from 0.
Is Support SingleRAN Mode	BTSCABINET.SRANMODE	Set this parameter based on site requirements.
Cabinet type	BTSCABINET.TYPE	Set this parameter based on the cabinet type. When there is no corresponding cabinet type, set this parameter to VIRTUAL .

Parameter Name	Parameter ID	Setting Notes
Cabinet Description	BTSCABINET.CABINETDESC	This parameter provides the user-defined description of the cabinet, including functions, location, and other information.
Is Master TMU Configured	BTSCABINET.ISMAINCABINET	This parameter is configurable only when the base station supports normalized data configuration. A cabinet with a BBU is defined as a basic cabinet, and a cabinet without a BBU is defined as an extension cabinet.
Master TMU Subrack No.	BTSCABINET.MAINCABINETSRN	This parameter is configurable only when the base station supports normalized data configuration.

Scenario 2: Adding BBU Boards

Table 3-85 describes the parameters for adding BBU boards.

Table 3-85 Parameters for adding BBU boards to GBTSs

Parameter Name	Parameter ID	Setting Notes
Index Type	BTSBRD.IDTYPE	Set this parameter according to site requirements.
BTS Index	BTSBRD.BTSID	None
BTS Name	BTSBRD.BTSNAME	None
Cabinet No.	BTSBRD.CN	The cabinet number starts from 0.
Subrack No.	BTSBRD.SRN	For a single-mode scenario: 0 For a multimode scenario: 0. If the value 0 is used, the recommended value is 1.
Slot No.	BTSBRD.SN	None
Board Type	BTSBRD.BT	Set this parameter based on site requirements.

3.4.1.2 Using MML Commands

Before using MML commands, refer to **3.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Adding Cabinets

Run the **ADD BTSCABINET** command to add a GBTS cabinet.

Adding BBU Boards

Run the **ADD BTSBRD** command to add boards to a GBTS cabinet.

3.4.1.3 Using the MAE-Deployment

To add cabinets and BBU boards on the MAE-Deployment, perform the following steps:

- Configuring a single base station
Follow the instructions in the following reference document and refer to [3.5.1.1 Data Preparation](#) to complete the initial configuration.
For details, see *GBTS Initial Configuration Guide (MAE-Deployment-based)* and navigate in the following sequence: **Creating a Single GBTS > Configuring GBTS Device Data > Configuring Cabinet, Subrack, and BBU Data > Procedure**.
- Configuring base stations in batches
You can save the data of a single base station (including data of the subrack and cabinet) as a template. Then, configure multiple base stations in batches by referencing this template. Follow the instructions in the following reference document to complete the initial configuration.
For details, see "Creating GBTSs in Batches" in *GBTS Initial Configuration Guide (MAE-Deployment-based)*.

3.4.2 Activation Verification

Run the **DSP BTSBRD** command to check whether the boards are running properly.

3.4.3 Network Monitoring

None

3.4.4 Reconfiguration

For the information on how to modify cabinets, subracks, and BBU boards on the MAE-Deployment, see "Introduction to the GBTS Device Panel" in the MAE-Access online help.

3.4.4.1 Modifying Cabinet Configuration

For GBTS cabinets, you can only modify the cabinet description.

Run the **MOD BTSCABINET** command to modify the parameters described in [Table 3-86](#).

Table 3-86 Parameters to be modified

Parameter Name	Parameter ID	Setting Notes
Index Type	BTSCABINET.IDTYPE	This parameter setting cannot be modified.
BTS Index	BTSCABINET.BTSID	This parameter setting cannot be modified.
BTS Name	BTSCABINET.BTSNAME	This parameter setting cannot be modified.
Cabinet No.	BTSCABINET.CN	This parameter setting cannot be modified.
Cabinet Description	BTSCABINET.CABINETDESC	This parameter gives the description of the cabinet.

3.4.4.2 Deleting BBU Boards

You can delete a board or module that is not required. Note that only a configured board or module can be deleted. Before deleting a board, delete the MOs associated with the board.

Run the **RMV BTSBRD** command to delete a board. [Table 3-87](#) describes the related parameters.

Table 3-87 Parameters for deleting a BBU board from a GBTS

Parameter Name	Parameter ID	Setting Notes
Index Type	BTSBRD.IDTYPE	Set this parameter to BYNAME or BYID based on site requirements.
BTS Index	BTSBRD.BTSID	This parameter uniquely identifies a base station under a BSC.
BTS Name	BTSBRD.BTSNAME	This parameter uniquely identifies a base station under a BSC.
Cabinet No.	BTSBRD.CN	Set this parameter to the number of the cabinet housing the board to be deleted.

Parameter Name	Parameter ID	Setting Notes
Subrack No.	BTSBRD.SRN	Set this parameter to the number of the subrack housing the board to be deleted.
Slot No.	BTSBRD.SN	Set this parameter to the number of the slot housing the board to be deleted.

3.4.4.3 Resetting BBU Boards

Run the **RST BTSBRD** command to reset a specified BBU board in a base station.

This command allows users to reset a board by means of software or hardware reset, which is specified by a parameter. If hardware reset is selected for a multimode base station, boards in the peer BBU can be reset.

Only normal boards support software reset. Normal boards are those BBU boards excluding the GTMU and UBRI.

The boards that support hardware reset are as follows:

- GTMU
- UBRI

3.4.4.4 Querying BBU Board Information

Run the **DSP BTSBRD** command to query the running status of a board.

3.5 Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB/Co-MPT Base Station)

3.5.1 Data Configuration

3.5.1.1 Data Preparation

Scenario 1: Adding Cabinets

[Table 3-88](#) describes the parameters for adding cabinets.

Table 3-88 Parameters for adding cabinets to eGBTs, NodeBs, eNodeBs, gNodeBs, or co-MPT base stations

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	CABINET.CN (<i>LTE eNodeB, 5G gNodeB</i>)	The cabinet number starts from 0.
Cabinet Type	CABINET.TYPE (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter is set according to the cabinet in use. When there is no corresponding cabinet type, this parameter is set to VIRTUAL .
Cabinet Description	CABINET.DESC (<i>LTE eNodeB, 5G gNodeB</i>)	None
Location Name	CABINET.LOCATIONNAME (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter is mandatory when BBUs are stacked in the centralized equipment room. The value of this parameter must be the same as that of LOCATION.LOCATIONNAME (<i>5G gNodeB, LTE eNodeB</i>). Both of them must be set to the name of the centralized equipment room, and LOCATION.LOCATIONNAME (<i>5G gNodeB, LTE eNodeB</i>) must be preferentially configured.
Location Name	LOCATION.LOCATIONNAME (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter is mandatory when BBUs are stacked in the centralized equipment room. You are advised to set this parameter to the name of the equipment room where the BBU subrack is located.
Location Type	LOCATION.LOCATIONTYPE (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter is mandatory when BBUs are stacked in the centralized equipment room. The recommended value is BBSITE .
Mode	LOCATION.MODE (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter is mandatory when BBUs are stacked in the centralized equipment room. If GNSS antennas are available in the centralized equipment room, you are advised to set this parameter to AUTO . Otherwise, this parameter is set to MANUAL .

Parameter Name	Parameter ID	Setting Notes
Longitude With Degree Format	LOCATION.LONGITUDEDEGFORMAT (LTE eNodeB, 5G gNodeB)	This parameter is mandatory when BBU's are stacked in the centralized equipment room. If LOCATION.MODE (LTE eNodeB, 5G gNodeB) is set to MANUAL , manual configuration is required. Otherwise, manual configuration is not required.
Latitude With Degree Format	LOCATION.LATITUDEDEGFORMAT (LTE eNodeB, 5G gNodeB)	This parameter is mandatory when BBU's are stacked in the centralized equipment room. If LOCATION.MODE (LTE eNodeB, 5G gNodeB) is set to MANUAL , manual configuration is required. Otherwise, manual configuration is not required.

Scenario 2: Adding Subracks

Subracks can be added to eGBTSS, NodeBs, eNodeBs, gNodeBs, and co-MPT base stations. [Table 3-89](#) describes the parameters for adding subracks.

Table 3-89 Parameters for adding subracks to eGBTSS, NodeBs, eNodeBs, gNodeBs, or co-MPT base stations

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	SUBRACK.CN	This parameter is set according to the cabinet in use.
Subrack No.	SUBRACK.SRN	This parameter is set according to the subrack in use. Subracks 0 and 1 can be BBU subracks, and subracks 4 and 5 can be RFU subracks.
Subrack Type	SUBRACK.TYPE	This parameter is set according to the subrack type in use.
Subrack Description	SUBRACK.DESC	None

Scenario 3: Adding BBU Boards

[Table 3-90](#) describes the parameters for adding BBU boards.

Table 3-90 Parameters for adding BBU boards to eGBTs, NodeBs, eNodeBs, gNodeBs, or co-MPT base stations

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	CN	The cabinet number starts from 0.
Subrack No.	SRN	For a single-mode scenario: 0 For a multimode scenario: 0. If subrack 0 is used, use subrack 1.
Slot No.	SN	None
Board Type	BT	This parameter is set according to the type of the board in use.
SubBoard Type	SBT	This parameter is set according to the type of the subboard in use. The UTRPc cannot be managed by GSM if GSM uses the GTMUc as its main control board in a NodeB/eNodeB/gNodeB/co-MPT base station.
Work Mode	BBP.WM	This parameter is set based on site requirements.
Hardware Capacity Enhance	BBP.HCE	- This HCE parameter can be set to PARTIAL only for the UBBP. - The HCE parameter can only be set to FULL for a UBBP when the BBWS parameter for the UBBP is set to a combination that includes UMTS.
Base Band Work Standard	BBP.BBWS	This BBWS parameter is supported only for the UBBP and UMDU. If the base station supports multimode concurrency, select multiple values from the corresponding drop-down list as required.

Parameter Name	Parameter ID	Setting Notes
Baseband Resource Alloc Mode	BBP.BbResAllocMode	- This parameter indicates the configuration method of baseband resource mode. It can be set to MANUAL(Manual) or AUTO(Auto). MANUAL(Manual) is the default value and is recommended. - This parameter can be configured only for UBBP series baseband processing units. The baseband processing units earlier than UBBPg series support only the manual mode. UBBPg series and later baseband processing units support both the manual mode and the auto mode. This parameter is an advanced parameter. To modify this parameter, contact Huawei Customer Service Center for technical support.
CPRI Extension	BBP.CPRIEX	This parameter is supported only for the LBBPd.
Board Specification	BBP.BRDSPEC	The value of this parameter must be consistent with the hardware type of the board. For example, set this parameter to UBBPd1 for a UBBPd1 and UBBPd2 for a UBBPd2. If the board specifications have to be forcibly verified using the MAE-Deployment, this parameter must be set to the hardware type of the board.
Cell Channel Number Extension	BBP.CCNE	This parameter is supported only for the UBBP and it takes effect only for the UBBPd9. In addition, the UBBPd9 will be reset after this parameter is modified. The parameter takes effect only in LTE.

Parameter Name	Parameter ID	Setting Notes
CPRI Interface Type	BBP.CPRIITFTYPE	For the UBBPg4, set this parameter to CPRI_QSFPPLUS when the DSFP optical module is used. For UBBPg boards other than the UBBPg1 and UBBPg4, set this parameter to CPRI_SFPPLUS when the DSFP optical module is used. For UBBPh boards, it is recommended that this parameter be set to CPRI_SFPPLUS when the SFP or DSFP optical module is used. If this parameter is set to CPRI_SFP or CPRI_6SFP_3SFPPLUS, the base station uses the value of CPRI_SFPPLUS.

NOTE

For details about the technical specifications of baseband processing units, see *Technical Specifications of Baseband Processing Units in 3900 & 5900 Series Base Station Product Documentation*.

If **BBP.BbResAllocMode** is set to **AUTO**, the following descriptions apply in capacity expansion scenarios:

- A change in site configuration data may cause a change in the settings of the service resource allocation parameters that the base station automatically selects for the baseband processing unit. To apply new settings, the baseband processing unit automatically resets. Note that the reset not only makes the new parameter settings take effect but also interrupts services carried in the corresponding cells.
- You are advised to plan the **STR BRDBBRESREALLOC** command for operation authorization. This command enables the baseband processing unit to automatically apply new settings of service resource allocation parameters, and is delivered after site configuration data is delivered.
- Before the base station obtains an operation authorization, the baseband processing unit continues to use the current settings of service resource allocation parameters. Certain cells activated on the baseband processing unit may fail to be established due to a service resource mismatch. As a result, these cells become unavailable.
- If the auto mode is planned for any baseband processing unit whose working RAT includes LTE and/or NR, all of the cells that work in the RAT in the base station must be bound to baseband equipment.

Scenario 4: Adding USU Boards

Table 3-91 describes the parameters for adding USU boards.

Table 3-91 Parameters for adding USU boards to eGBTs, NodeBs, eNodeBs, gNodeBs, or co-MPT base stations

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	<i>CN</i>	The cabinet number starts from 0.
Subrack No.	<i>SRN</i>	Use subrack 1 if subrack 0 in the same cabinet is used. Otherwise, use subrack 0.
Slot No.	<i>SN</i>	None
Board Type	<i>BT</i>	This parameter is set according to the type of the board in use.

3.5.1.2 Using MML Commands

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Adding Cabinets and Subracks

Run the **ADD CABINET** command to add a cabinet on the eGBTs/NodeB/eNodeB/gNodeB side. Run the **ADD SUBRACK** command to add a subrack on the eGBTs/NodeB/eNodeB/gNodeB side.

When the BBU subracks are stacked in the centralized equipment room, run the **ADD LOCATION** command to add the information about the equipment room before adding a cabinet. Then, run the **ADD CABINET** command to add a cabinet.

Adding BBU/USU Boards

Run the **ADD BRD** command to add boards to a cabinet on the eGBTs/NodeB/eNodeB/gNodeB side.

NOTE

The BTS5900R is an integrated device. Therefore, you do not need to configure cabinets, subracks, or boards.

3.5.1.3 Using the MAE-Deployment

To add cabinets, subracks, and BBU/USU boards on the MAE-Deployment, perform the following steps:

- Configuring a single base station on the GUI
Follow the instructions in the following reference documents and refer to [3.5.1.1 Data Preparation](#) to complete the initial configuration.
 - For details, see *NodeB Initial Configuration Guide (MAE-Deployment-based)* and navigate in the following sequence: **Creating a Single NodeB > Configuring NodeB Device Data > Configuring Cabinet, Subrack, and BBU Data > Procedure.**
 - For details, see *eNodeB Initial Configuration Guide (MAE-Deployment-based)* and navigate in the following sequence: **Creating a Single eNodeB > Configuring eNodeB Device Data > Configuring Cabinet, Subrack, and BBU Data > Procedure.**
 - For details, see *gNodeB Initial Configuration Guide (MAE-Deployment-based)* and navigate in the following sequence: **Creating a Single gNodeB > Configuring gNodeB Device Data > Configuring Cabinet, Subrack, and BBU Data > Procedure.**
 - For details, see *Co-MPT Base Station Initial Configuration Guide (MAE-Deployment-based)* and navigate in the following sequence: **Creating a Single Co-MPT Base Station > Configuring Device Data About the Co-MPT Base Station > Configuring Cabinet, Subrack, and BBU Data > Procedure.**
 - For details, see *Separate-MPT Base Station Initial Configuration Guide (MAE-Deployment-based)* and navigate in the following sequence: **Creating a Single Separate-MPT Base Station > Configuring Device Data About the Separate-MPT Base Station > Configuring Cabinet, Subrack, and BBU Data > Procedure.**
- Configuring base stations in batches
You can save the data of a single base station (including data of the subrack and cabinet) as a template. Then, configure multiple base stations in batches by referencing this template. Follow the instructions in the following reference documents to complete the initial configuration.
 - For details, see "Creating NodeBs in Batches (MOC Export)" or "Creating NodeBs in Batches (Customization Tool)" in *NodeB Initial Configuration Guide (MAE-Deployment-based)*.
 - For details, see "Creating eNodeBs in Batches" in *eNodeB Initial Configuration Guide (MAE-Deployment-based)*.
 - For details, see "Creating gNodeBs in Batches" in *gNodeB Initial Configuration Guide (MAE-Deployment-based)*.
 - For details, see "Creating Co-MPT Base Stations in Batches (MOC Export)" or "Creating Co-MPT Base Stations in Batches (Customization Tool)" in *Co-MPT Base Station Initial Configuration Guide (MAE-Deployment-based)*.
 - For details, see "Creating Separate-MPT Base Stations in Batches" in *Separate-MPT Base Station Initial Configuration Guide (MAE-Deployment-based)*.

3.5.2 Activation Verification

Run the **DSP BRD** command to check whether the boards are running properly.

3.5.3 Network Monitoring

None

3.5.4 Reconfiguration

For the information on how to modify cabinets, subracks, and BBU boards on the MAE-Deployment, see "Introduction to the Co-MPT Base Station Device Panel" in the MAE-Access online help.

3.5.4.1 Modifying Cabinet Configuration

For eGBTS, NodeB, eNodeB, gNodeB, and co-MPT base station cabinets, you can modify the cabinet description and change the cabinet type in specific scenarios.

You can run the **MOD CABINET** command to modify related parameters when the BBU subrack is not moved to another centralized equipment room, as shown in [Table 3-92](#).

- You can only change the virtual cabinet to other types of cabinets.
- If a cabinet to be modified contains non-BBU subracks, you cannot change the cabinet type.
- If a cabinet to be modified contains only BBU subracks and can house additional BBU subracks, you can change the cabinet type. Otherwise, you cannot change the cabinet type.
- If a cabinet to be modified contains no subracks, you can change the cabinet type according to the onsite cabinet type.

Table 3-92 Parameters to be modified

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	CABINET.CN	This parameter setting cannot be modified.
Cabinet Description	CABINET.DESC	This parameter provides the user-defined description of the cabinet, including functions, location, and other information.
Cabinet Type	CABINET.TYPE	This parameter is set based on the cabinet type. When there is no corresponding cabinet type, this parameter is set to VIRTUAL .

When the BBU subrack is moved to another centralized equipment room, run the following commands to configure related parameters. For details about how to modify parameters, see [Table 3-93](#).

1. Run the **RMV LOCATION** command to delete the information about the source equipment room.
2. Run the **ADD LOCATION** command to add information about the target equipment room.
3. Run the **MOD CABINET** command to modify parameters related to the cabinet where the BBU is located.

Table 3-93 Parameters for moving a BBU subrack to another centralized equipment room

Parameter Name	Parameter ID	Setting Notes
Location Name	LOCATION.LOCATIONNAME (<i>LTE eNodeB, 5G gNodeB</i>)	When the BBU is moved to another centralized equipment room, you are advised to set this parameter to the name of the target equipment room.
Location Type	LOCATION.LOCATIONTYPE (<i>LTE eNodeB, 5G gNodeB</i>)	When the BBU is moved to another centralized equipment room, you are advised to configure the BBU based on the location type of the target equipment room.
Mode	LOCATION.MODE (<i>LTE eNodeB, 5G gNodeB</i>)	When the BBU is moved to another centralized equipment room, you are advised to configure the BBU based on the mode of the target equipment room. If GNSS antennas are available in the target equipment room, you are advised to set this parameter to AUTO . Otherwise, this parameter is set to MANUAL .
Longitude With Degree Format	LOCATION.LONGITUDEDEGREEFORMAT (<i>LTE eNodeB, 5G gNodeB</i>)	When the BBU is moved to another centralized equipment room, you are advised to configure the BBU based on the longitude of the target equipment room. If LOCATION.MODE (<i>LTE eNodeB, 5G gNodeB</i>) is set to MANUAL , manual configuration is required. Otherwise, manual configuration is not required.

Parameter Name	Parameter ID	Setting Notes
Latitude With Degree Format	LOCATION.LATITUDEDEGFORMAT (LTE eNodeB, 5G gNodeB)	When the BBU is moved to another centralized equipment room, you are advised to configure the BBU based on the latitude of the target equipment room. If LOCATION.MODE (LTE eNodeB, 5G gNodeB) is set to MANUAL , manual configuration is required. Otherwise, manual configuration is not required.
Cabinet No.	CABINET.CN	This parameter setting cannot be modified.
Cabinet Description	CABINET.DESC	This parameter provides the user-defined description of the cabinet, including functions, location, and other information.
Cabinet Type	CABINET.TYPE	This parameter is set based on the cabinet type. When there is no corresponding cabinet type, this parameter is set to VIRTUAL .
Location Name	CABINET.LOCATIONNAME	When the BBU is moved to another centralized equipment room, you are advised to set this parameter based on the actual name of the target equipment room. You need to modify the location name of the cabinet where this BBU is located to the name of the target equipment room.

3.5.4.2 Modifying Subrack Configuration

You can modify the subrack type and subrack description on an eGBTS, NodeB, eNodeB, gNodeB, or co-MPT base station.

Run the **MOD SUBRACK** command to modify the parameters described in [Table 3-94](#) on an eGBTS, NodeB, eNodeB, gNodeB, or co-MPT base station.

Table 3-94 Parameters to be modified

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	SUBRACK.CN	This parameter setting cannot be modified.
Subrack No.	SUBRACK.SRN	This parameter setting cannot be modified.
Subrack Type	SUBRACK.TYPE	Set this parameter to BBU3900, BBU3910, BBU5910 or BBU5900 based on site requirements.
Subrack Description	SUBRACK.DESC	This parameter indicates the description of the subrack, such as function and location.

3.5.4.3 Modifying BBU Board Configuration

You can modify the BBP and UTRP configurations of eGBTs, NodeBs, eNodeBs, and co-MPT base stations, as well as the BBP configurations of gNodeBs.

Modifying the BBP Configuration of eGBTs, NodeBs, eNodeBs, gNodeBs, or Co-MPT Base Stations

Run the **MOD BBP** command to modify the BBP configuration, as described in [Table 3-95](#).

- Modifying the value of the **Board Type**, **Work Mode**, **Hardware Capacity Enhance**, or **CPRI Extension** parameter interrupts services on the BBP.
- After the BBP (LBBP/UBBP) resets, services on this board are restored.

Pay attention to the following items when modifying the LBBP configuration:

- If the **RRUCHAIN** MO is configured on any of optical ports 3 through 5 on an LBBP, the **CPRIEX** parameter cannot be set to **ON**.
- Before modifying the working mode of a BBP serving activated cells, ensure that after the modification, the eNodeB or gNodeB is configured with a board that supports those activated cells and has sufficient idle resources (see the cell specifications of the BBP). Otherwise, the cells cannot be activated.
- BBPs in FDD support 1T1R, 1T2R, 2T2R, 2T4R, and 4T4R FDD cells.
- BBPs in TDD support 1T1R, 2T2R, 4T4R, and 8T8R TDD cells.
- If the **RRUCHAIN** MO in load sharing mode or ring topology is configured on BBPs and the chain head and tail belong to different BBPs, the working mode of the BBP cannot be modified.
- If the RRU/RFU cannot set up links or work at the maximum line rate after the BBP is blocked/unblocked, run the **STR CPRI LBRNEG** command to manually initiate negotiation.

- After the board is blocked, the Board Blocked alarm will be reported and the board will not provide services. After the board is unblocked, the alarm will be cleared.

Table 3-95 Modifying the BBP configuration of eGBTs, NodeBs, eNodeBs, gNodeBs, or co-MPT base stations

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	BBP.CN	This parameter setting cannot be modified.
Subrack No.	BBP.SRN	This parameter setting cannot be modified.
Slot No.	BBP.SN	This parameter setting cannot be modified.
Work Mode	BBP.WM	This parameter is set based on site requirements.
Hardware Capacity Enhance	BBP.HCE	• This HCE parameter can be set to PARTIAL only for the UBBP. • The HCE parameter can only be set to FULL for a UBBP when the BBWS parameter for the UBBP is set to a combination that includes UMTS.
CPRI Extension	BBP.CPRIEX	This parameter is supported only for the LBBPd.
Base Band Work Standard	BBP.BBWS	This BBWS parameter is supported only for the UBBP and UMDU. Before changing a UBBP's working mode, terminate the services that are being carried by the UBBP.

Parameter Name	Parameter ID	Setting Notes
Baseband Resource Alloc Mode	BBP.BbResAllocMode	- This parameter indicates the configuration method of baseband resource mode. It can be set to MANUAL(Manual) or AUTO(Auto). MANUAL(Manual) is the default value and is recommended. - This parameter can be configured only for UBBP series baseband processing units. The baseband processing units earlier than UBBPg series support only the manual mode. UBBPg series and later baseband processing units support both the manual mode and the auto mode. This parameter is an advanced parameter. To modify this parameter, contact Huawei Customer Service Center for technical support.
Board Specification	BBP.BRDSPEC	The value of this parameter must be consistent with the hardware type of the board. For example, set this parameter to UBBPd1 for a UBBPd1 and UBBPd2 for a UBBPd2. If the board specifications have to be forcibly verified using the MAE-Deployment, this parameter must be set to the hardware type of the board.
Cell Channel Number Extension	BBP.CCNE	This parameter is supported only for the UBBP and it takes effect only for the UBBPd9. In addition, the UBBPd9 will be reset after this parameter is modified. The parameter takes effect only in LTE.

Parameter Name	Parameter ID	Setting Notes
CPRI Panel Port Number	BBP.CPRIPANELPORTNUM	This parameter can be set only for the UBBP board. The number of ports on the CPRI panel must be the same as the number of CPRI ports provided by the board. Otherwise, the board cannot be started. When the UBBP board provides nine ports on the fronthaul panel, this parameter must be set to 9; when it provides only six ports on the fronthaul panel, the parameter must be set to 6. If this parameter is modified using the MAE-Deployment, Port Usage (specified by CASCADEPORT.PORTUSAGE (LTE eNodeB, 5G gNodeB)) of the cascade port corresponding to the board must be changed accordingly. • If the number of ports on the CPRI panel is changed from 9 to 6, Port Usage of cascade port 6 must be set to INTERCONNECT. • If the number of ports on the CPRI panel is changed from 6 to 9, Port Usage of cascade port 6 must be set to FRONTHAUL.

Modifying Transmission Board Configuration of eGBTs, NodeBs, eNodeBs, or Co-MPT Base Stations

Run the **MOD UTRP** command to modify the subboard type of the UTRP. [Table 3-96](#) describes the related parameters.

Table 3-96 Modifying transmission board configuration of eGBTs, NodeBs, eNodeBs, or co-MPT base stations

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	TRP.CN	This parameter setting cannot be modified.

Parameter Name	Parameter ID	Setting Notes
Subrack No.	TRP.SRN	This parameter setting cannot be modified.
Slot No.	TRP.SN	This parameter setting cannot be modified.
SubBoard Type	TRP.SUBTYPE	This parameter is set based on site requirements. The UTRPc cannot be managed by GSM if GSM uses the GTMUC as its main control board in a NodeB/eNodeB/gNodeB/co-MPT base station.

3.5.4.4 Deleting BBU Boards

You can delete a board or module that is not required. Note that only a configured board or module can be deleted. Before deleting a board, delete the MOs associated with the board.

Run the **RMV BRD** command to delete a BBU board. [Table 3-97](#) describes the related parameters.

- Deleting a board causes the board to reset.
- After a board is deleted, all services carried by the board are interrupted.
- If a BBP has been configured with the **RRUCHAIN** MO, the board cannot be deleted. If you want to delete the board, delete the **RRUCHAIN** MO first.
- If a board configured with a subboard is deleted, the subboard is deleted accordingly.

Table 3-97 Parameters for deleting a BBU board from an eGBTS, NodeB, eNodeB, gNodeB, or co-MPT base station

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	CN	Set this parameter to the number of the cabinet housing the board to be deleted.
Subrack No.	SRN	Set this parameter to the number of the subrack housing the board to be deleted.
Slot No.	SN	Set this parameter to the number of the slot housing the board to be deleted.

3.5.4.5 BBU Board Maintenance

3.5.4.5.1 Resetting BBU Boards

Resetting BBU Boards in Soft Reset Mode in eGBTs, NodeBs, eNodeBs, gNodeBs, or Co-MPT Base Stations

- If a board has a fault that cannot be rectified, soft reset is required. Soft reset takes a relatively short time.

Run the **RST BRD** command to perform a soft reset.

NOTE

USCU, FAN, UPEU, and UEIU do not support soft reset.

The BTS5900R is integrated equipment. Therefore, running the **RST BRD** command resets the entire base station.

After the **RST BRD** command is executed, the BBU Board Maintenance Link Failure alarm and the Board Startup event are generated. Run the **DSP BRD** command to check the values of **Availability Status** and **Operational State**. If the value of **Availability Status** is **Normal** and the value of **Operational State** is **Enabled**, the board has started up properly and the fault has been rectified.

- For the UBBPh3a and UBBPh3b, the base station reports ALM-26234 BBU CPRI Interface Error in either of the following two scenarios. In this case, you can reset the board to make the CPRI interface configuration take effect by running the **RST BRD** command or enabling the function of automatic reset upon CPRI interface resource conflicts before the switchover. The **CPRI_CONFLICT_AUTO_RST_SW** option of the **EQUIPMENT.EQMOPTFEATSW (LTE eNodeB, 5G gNodeB)** parameter specifies whether to enable automatic reset upon CPRI interface resource conflicts. You can run the **SET EQUIPMENT** command to configure related parameters to enable this function.
 - Scenario 1: The BBP is switched from not using the DSFP module to using it and is configured with an RRU chain/ring on subport 1.

NOTE

In this scenario, if the board is not reset, the networking information of sub-port 1 of the DSFP optical module cannot be queried using **DSP RRUCHAINPHYTOPO**.

- Scenario 2: The BBP is switched from using the DSFP module to not using it and the RRUs connected to ports 6 to 8 on the BBP are configured with GSM cell-related settings.

Resetting BBU Boards in Power-off Reset Mode in eGBTs, NodeBs, eNodeBs, gNodeBs, or Co-MPT Base Stations

If soft reset fails, power-off reset is required. Power-off reset takes a relatively long time.

Run the **RST BRDPWROFF** command to perform a power-off reset.

NOTE

The FAN, UPEU, and UEIU do not support power-off reset.

The BTS5900R is integrated equipment. Therefore, running the **RST BRDPWROFF** command resets the entire base station.

After the **RST BRDPWROFF** command is executed, the BBU Board Maintenance Link Failure alarm and the Board Startup event are generated. Run the **DSP BRD** command to check the values of **Availability Status** and **Operational State**. If the value of **Availability Status** is **Normal** and the value of **Operational State** is **Enabled**, the board has started up properly and the fault has been rectified.

3.5.4.5.2 Querying BBU Board Information

- Step 1** Run the **DSP BRD** command to query the running status of a board.
- Step 2** Run the **DSP BRDMFRINFO** command to query the manufacturing information about a board.

----End

4 Glossary

There are no glossary associated with this feature.

5 Reference Documents

1. *DBS3900 Installation Guide*
2. *BTS3900 (Ver.C) Installation Guide*
3. *BTS3900A (Ver.C) Installation Guide*
4. *BTS3900L (Ver.C) Installation Guide*
5. *BTS3900AL (Ver.A) Installation Guide*
6. *BTS3900C WCDMA Installation Guide*
7. *BTS3900C (Ver.C) Installation Guide*
8. *BTS3900 (Ver.D) Installation Guide*
9. *BTS3900A (Ver.D) Installation Guide*
10. *BTS3900L (Ver.D) Installation Guide*
11. *BTS3900 Installation Guide for Enhancing Existing Cabinets*
12. *BTS3900A Installation Guide for Enhancing Existing Cabinets*
13. *BTS3900L Installation Guide for Enhancing Existing Cabinets*
14. *DBS3900 Installation Guide for Enhancing Existing Cabinets*
15. *BSC6910 GSM MML Command Reference*
16. *BSC6900 GSM MML Command Reference*
17. *BSC6910 GSM Parameter Reference*
18. *BSC6900 GSM Parameter Reference*
19. *BSC6910 GSM Alarm Reference*
20. *BSC6900 GSM Alarm Reference*
21. *BSC6910 GSM Event Reference*
22. *BSC6900 GSM Event Reference*
23. *3900 & 5900 Series Base Station MML Command Reference*
24. *3900 & 5900 Series Base Station Parameter Reference*
25. *3900 & 5900 Series Base Station Alarm Reference*
26. *3900 & 5900 Series Base Station Event Reference*
27. *3900 & 5900 Series Base Station Initial Configuration Guide*
28. *BTS3900 Installation Guide for Enhancing Existing Cabinets (3900 & 5900 Series)*

29. *BTS3900L Installation Guide for Enhancing Existing Cabinets (3900 & 5900 Series)*
30. *DBS5900 Installation Guide*
31. *Technical Specifications of Baseband Processing Units*